The Null Zoo: Size and Power of Backtest-Overfitting and Data-Snooping Tests on Synthetic Searches with Known Ground Truth

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Abstract

Corrections for backtest overfitting and data snooping—the deflated Sharpe ratio, the multiple-testing haircut, family-wise adjustments, White's Reality Check and Hansen's test for superior predictive ability (SPA)—are derived under assumptions that trading research violates. The Null Zoo is an open, versioned, pre-registered simulation benchmark that draws research searches (20 strategies, 504 daily returns) from nine return families with known ground truth and scores 14 validators and an oracle by size and power, with 10,000 searches per cell. With a stationary bootstrap of mean block length 8, Hansen's studentized SPA test and the stepwise SPA test built on it rejected more than 6% of skill-less searches at a nominal 5% in at least seven of nine families in each of two runs (up to 12.6%); the Reality Check exceeded 6% only under autocorrelation. The bootstrap holds fixed a noisy, downward-biased long-run variance; a closed-form approximation, with proofs, shows that the resulting size exceeds the nominal level and grows with the block length, and accounts for about nine tenths of the excess. Re-studentizing every resample, as Romano and Wolf recommend, brought the size to about the Reality Check's (5.0–6.2% outside the autocorrelated family). The released SPA test of the Python package arch does not studentize; a change merged on 6 October 2026 adopted the fixed-variance version, which rejected 13.5% of i.i.d. normal and 19.1% of negatively skewed skill-less searches at its defaults, and after our report arch's maintainer replaced it with a re-studentizing version that rejected 3.7–6.7% across the nine families in a pre-registered confirmation on fresh seeds. On 162–207 published cross-sectional predictors in 9 ten-year windows, the fixed-variance stepwise test declared 32% more predictors significant than a bootstrap-t, and 55% more at arch's default block length. A lag-one autocorrelation of 0.2 raises the size of single-series tests that ignore it to about 19.5%, and one of 0.05 to about 7.55%, as a closed form predicts for independent and for equicorrelated strategies; and the deflated Sharpe ratio's 0.95 rule misses 90.8% of i.i.d. searches containing a strategy with a true Sharpe ratio of 2. Of 34 pre-registered predictions and hypotheses, 30 held.

1 Introduction

A researcher who tries many variants of a trading strategy and reports the best one reports a biased number. The best of N strategies with no skill has an in-sample Sharpe ratio that grows with N, roughly in proportion to $\sqrt{2\ln N}$ times the spread of the individual estimates [4], and the same selection effect pervades published factor research [30]. Two families of corrections are now standard. The first adjusts the best strategy's own statistic: the deflated Sharpe ratio [3], built on the probabilistic Sharpe ratio [2]; the multiple-testing haircut of Harvey and Liu [28]; and

family-wise adjustments such as those of Holm [32] and Šidák [49]. The second resamples the whole search: White's Reality Check [56], applied to trading rules by Sullivan et al. [52], Hansen's test for superior predictive ability [27] and the stepwise procedures of Romano and Wolf [45] and Hsu et al. [33].

Each correction rests on assumptions—normal or independent returns, independent trials, large samples—that real searches violate in predictable ways. Returns are fat-tailed, skewed, volatility-clustered and sometimes autocorrelated; strategy variants share most of their positions and so are correlated; and two years of daily data is a short sample for a variance estimate. Inference for a single Sharpe ratio under such violations is well studied [36, 37, 41], and simulations have assessed multiple-testing corrections for the cross-section of factors and anomalies [14, 29]. We are not aware of a tabulation of how much each violation distorts each *selection* correction when it is applied, as practitioners apply it, to the best of one strategy search.

This paper provides one with a simulation benchmark, the *Null Zoo*. It draws complete research searches—20 candidate strategies of 504 daily returns each—from nine return families whose ground truth is known by construction, applies 14 validators and an oracle as their implementations are meant to be called, and records how often each declares a skilled strategy when there is none (its *size*) and when one strategy has a true annualized Sharpe ratio of 1, 2 or 3 (its *power*). The main run has 10,000 searches per cell. Every governed run was pre-registered before it was run, following recent arguments that comparative simulation studies need the same protection against selective reporting as empirical ones [42, 50]. A surprising result was confirmed on fresh seeds under a new pre-registration before it was used; its mechanism, suggested by an unregistered pilot, was confirmed in a registered run of a second implementation in another language; a further registered run maps where the remedies work and applies the tests to real data; and a fifth re-tests the newest, initially unregistered, checks on fresh seeds.

The findings, with a stationary bootstrap of mean block length round$(T^{1/3}) = 8$ unless stated:

1. **Hansen's studentized SPA test, and the stepwise SPA test built on it, are oversized at this horizon; White's Reality Check is close to its level except under autocorrelation.** At a nominal 5%, the consistent SPA test rejected more than 6% of skill-less searches in 7 of the 9 families in one run and in 8 in its confirmation, up to 12.6%; the Reality Check rejected 4.5–5.7% outside the autocorrelated family and 6.3–6.4% within it.

2. **The cause is a variance estimate the bootstrap holds fixed.** The statistic divides each mean by a long-run standard deviation estimated once and reused in every resample. That estimate is noisy and, computed from demeaned data, biased downward. Proposition 1 gives the resulting size in closed form under i.i.d. normal returns, proves that it exceeds the nominal level and grows with the block length, and accounts for 89% and 91% of the measured excess at block lengths 8 and 22. Under negative skew a mean and its estimated standard deviation are also dependent. Romano and Wolf [45] note that holding the estimate fixed is only first-order accurate and recommend re-estimating it in every resample; doing so with the resample's standard deviation brought the size to about the Reality Check's (5.0–6.2% outside the autocorrelated family). The excess shrinks with the sample, grows with the number of strategies and the block length, and reaches 12.9% (17.8% under negative skew) at the default block length of the Python package arch; with the i.i.d. resampling of Hansen's own simulations it nearly disappears.

3. **Long blocks make every block-bootstrap test liberal,** because the bootstrap's conditional variance of a mean is the same estimate: at block length 22 the Reality Check and the bootstrap-t were liberal with up to 20 strategies.

4. **The fix holds in code we did not write.** In arch 8.0.0 [48], the latest release, the `studentize` option has no effect. A change merged on 6 October 2026 [55] made the default the fixed-variance test, which, run directly, rejected 13.5% of i.i.d. normal and 19.1% of negatively skewed null searches at its default block length. After we reported this [10], the maintainer merged a change the same day [47] that re-studentizes every resample by its own long-run variance; on 2,000 new null searches per family it rejected 3.8–6.4% at its default block length, against 8.5–17.8% for the fixed-variance test on the same searches. Correcting the author's own implementation the same way took the consistent SPA test from 12.0% to 5.2% under negative skew and from 7.2% to 5.3% under i.i.d. normal returns. A pre-registered confirmation on fresh seeds reproduced both results: arch's corrected test rejected 3.7–6.7% at its default block length across the nine families, against 7.7–18.2% for the fixed-variance test on the same searches.

5. **Studentizing and recentring pay where they were designed to:** studentizing when the skilled strategy is among the less volatile (power 5.4% to 53.6%), consistent recentring when the other strategies are clearly poor, provided ties are counted.

6. **Autocorrelation breaks the single-series tests that ignore it, and little is needed.** A lag-one autocorrelation of 0.2 raises their false-positive rate to about 19.5%, and one of 0.05, which makes the i.i.d. standard error only 5.1% too small, raises the one-sided haircut's to 7.6%. Proposition 2 predicts these sizes in closed form by inserting Lo's variance inflation into the best-of-N tail: the far-tail cutoff of a best-of-20 test makes its size about 8.4 times as sensitive, in proportion, to an error in the standard error, against 3.4 for a single test. Part (b) of the proposition extends the closed form to equicorrelated strategies: correlation makes the tests conservative but leaves them almost as sensitive. Lo's factor removes most of the excess in this, its best case, kernel and block-bootstrap remedies for a single series reduce it, and the joint block-bootstrap tests, re-studentized or unstudentized, stayed close to their level at these autocorrelations. A pre-registered confirmation reproduced these sizes and both parts of the proposition; it also found LET with Lo's factor about half a point liberal.

7. **The deflated Sharpe ratio at its authors' 0.95 rule tests a harder hypothesis than "no skill",** and fails to accept 90.8% of i.i.d. searches containing a strategy with a true Sharpe ratio of 2 over two years.

8. **Selection, not correction, sets the ceiling.** An oracle that knows which strategy is skilled rejects in 87.7% of such searches; the best calibrated best-of-20 tests reach about 52.9%.

9. **On real data the choice matters at the margin.** Applied to 20 moving-average timing rules in 51 two-year windows of US market data since 1927, the tests disagree in 5 windows, and the tests found oversized reject most often. On published cross-sectional predictors, in a pre-registered illustration, the fixed-variance stepwise test declared 32% more predictors significant than the bootstrap-t, and 55% more at arch's default block length.

The mechanism in finding 2—a noisy long-run variance inflating a studentized test—is familiar from heteroskedasticity- and autocorrelation-robust inference [34, 35, 53] and from tests of equal forecast accuracy [15, 17, 31], and Romano and Wolf state the remedy for data-snooping tests. What the benchmark adds is the size of the effect where strategy research operates, its interaction with the maximum over strategies, with skewness and with the block length, evidence for Romano and Wolf's remedy and for its limits under serial dependence, and its consequences for widely used code, where it has already led to a correction.

The benchmark scores a statistic introduced by its own author, the *luck-equivalent-trials* count, and the v1 and v1b runs chose the default tests of validation software published by the author's firm: the pre-registered rules chose the Reality Check for searches and LET-Lo, the author's statistic with Lo's factor, for single series. Bartoš et al. [6] argue that simulation studies designed by a method's developers carry a conflict of interest. We report that conflict (see Competing interests), pre-register the runs, publish every generator, validator, seed and result file, and note that the benchmark's clearest failures—autocorrelation, and the floored strict-inequality p-value—apply to the author's own implementations as much as to anyone's.

Related work. Arian et al. [1] compare out-of-sample testing schemes—walk-forward, k-fold, purged and combinatorial purged cross-validation—in a synthetic market with known structure, scoring them by the probability of backtest overfitting [5] and the deflated Sharpe ratio. Their object is the scheme that produces a backtest; ours is the correction applied to its result. Santoni et al. [46] combine the deflated Sharpe ratio, the probability of backtest overfitting, the SPA test and the minimum track-record length into one robustness grade, and report its discrimination on synthetic markets and a pre-registered test on real ones; our results bear on its SPA component, whose size depends on how it is studentized and on the block length. Harvey and Liu [29] and Chordia et al. [14] study false and missed discoveries among many factors or anomalies, where the tests are applied across a cross-section rather than to the best of one search. Hansen [27], Romano and Wolf [45] and Hsu et al. [33] evaluate their tests by simulation in their own designs; Section 4.3 reconciles their results with ours. The size and power of multiple-testing procedures under dependence are well studied in statistics; our contribution is a small, open, versioned and pre-registered benchmark that uses the return shapes and search structures of trading research and calls each correction the way it is called in practice.

2 Design

We plan and report the study with the ADEMP structure of Morris et al. [39]: aims, data-generating mechanisms, estimands, methods and performance measures.

2.1 Aims

For each validator and each return family the study estimates (i) the probability that the validator calls the best of N skill-less strategies skilled (its *size*), and (ii) the probability that it rejects when one of the N strategies has real skill (its *power*). A validator is *calibrated* in a family when its size is within Monte Carlo error of its nominal level.

2.2 Data-generating mechanisms

One replication draws a *search*: $N = 20$ return series of $T = 504$ periods, read as two years of daily returns (252 periods per year). Every series has population mean zero and population variance one, so every strategy's true Sharpe ratio is zero. In the skill arms, series 0 receives a constant drift of $s/\sqrt{252}$ per period, which gives it a true annualized Sharpe ratio of $s \in \{1, 2, 3\}$; the other series are unchanged. The nine families are:

IID normal. Independent standard normal returns; the case every correction is derived for.

Student t_4. Student t with 4 degrees of freedom, scaled to unit variance: symmetric fat tails. Its fourth moment is infinite, so sample kurtosis does not converge.

Skew ±1.32. A centred, unit-variance lognormal with log-standard deviation 0.4, whose skewness is +1.32, and its mirror image (−1.32).

GARCH(1,1). $h_t = 0.05 + 0.10\,\varepsilon_{t-1}^2 + 0.85\,h_{t-1}$, $\varepsilon_t = \sqrt{h_t}\,z_t$ [8], with unconditional variance one; each series starts at $h = 1$, $\varepsilon = 0$, without burn-in.

AR(1). $x_t = 0.2\,x_{t-1} + \sqrt{1 - 0.2^2}\,z_t$, started from its stationary distribution.

Two volatility regimes. Normal returns whose standard deviation is 0.50 or 1.32 (0.6 and 1.6 rescaled to unit variance), switching with probability 0.02 each period.

Correlated trials. $x_{k,t} = \sqrt{1/2}\,f_t + \sqrt{1/2}\,z_{k,t}$ with a factor f_t common to all strategies, so every pair of strategies has correlation 0.5.

Block clusters. Four independent clusters of five strategies; strategies in a cluster share a factor with pairwise correlation 0.8. Twenty strategies that are really about four ideas, the shape a parameter grid around a few rules produces.

Instrument tests run before any validator is scored: every family's sample mean, variance, skewness, lag-one autocorrelation, autocorrelation of squared returns and cross-strategy correlation are checked against the property it claims, and the drift is checked to give series 0, and only series 0, the intended Sharpe ratio. Every null hypothesis in the nine families is the least favourable configuration, in which every strategy's mean is exactly zero; Section 4 adds designs with poor alternatives and unequal volatilities.

2.3 Estimands

In each replication the validator sees all 20 series and returns a p-value for the hypothesis that the best strategy has no skill; the single-series validators select the strategy with the highest sample Sharpe ratio, the joint tests consider all strategies at once. The estimands are the rejection rates $\Pr(p \le 0.05)$ in the null arm (size) and the skill arms (power). Because raw power rewards a liberal test, we also report size-adjusted power: the power at the largest attainable cutoff at which the same validator rejects at most 5% of the matching null cell. A user cannot apply that cutoff without knowing the family, but it shows how well each statistic ranks skilled searches above skill-less ones.

2.4 Methods

The benchmark scores 14 validators and an oracle. Let $\widehat{SR}$ be the best strategy's per-period Sharpe ratio (sample standard deviation with divisor $T - 1$), $\widehat{SR}_a = \widehat{SR}\sqrt{252}$ its annualized value, and $\hat{\gamma}_3, \hat{\gamma}_4$ its sample skewness and (non-excess) kurtosis. The single-series validators are:

LET. The luck-equivalent-trials test. Under the null and normal returns, $t = \widehat{SR}\sqrt{T}$ is Student t with $T - 1$ degrees of freedom, so one skill-less strategy reaches the observed value with probability $p_1 = \Pr(t_{T-1} \ge t)$, and the best of N independent ones with probability $1 - (1 - p_1)^N$ [49], the reported p-value. As a test it is the Šidák-adjusted Student t test; the statistic reports the same evidence in units of trials, $N_\pi = \ln(1-\pi)/\ln(1-p_1)$ for a chosen probability π.

LET-Lo. LET after deflating $\widehat{SR}_a$ by the autocorrelation-adjusted annualisation factor of Lo [37], $\left[1 + \frac{2\hat{\rho}}{1-\hat{\rho}}\left(1 - \frac{1-\hat{\rho}^{252}}{252(1-\hat{\rho})}\right)\right]^{-1/2}$, at the best series' lag-one autocorrelation $\hat{\rho}$ (clamped to $[-0.95, 0.95]$), as the haircut code of Harvey and Liu [28] does. The factor assumes $\rho_k = \rho^k$, which the AR(1) family satisfies exactly.

LET-NN. LET with the normal approximation and the non-normal standard error of Mertens [38] evaluated at the estimate: $z = \widehat{SR}\sqrt{T-1}\,/\sqrt{1-\hat{\gamma}_3\widehat{SR}+\frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}$.

HL, HL-Lo, Holm. The haircut Sharpe ratio of Harvey and Liu [28]: the best strategy's two-sided Student t p-value, adjusted by Bonferroni for 20 tests, without and with the Lo factor, and the Holm [32] variant. The Holm variant ranks the two-sided p-values of all 20 strategies, so it differs from Bonferroni only when a strategy with a strongly negative Sharpe ratio has the smallest two-sided p-value. Harvey and Liu define the haircut with two-sided p-values, which for a positive Sharpe ratio is a one-sided test at 2.5%; to compare tests at a common level we also read it one-sided (two-sided p halved), and the tables report that reading unless stated.

DSR. The deflated Sharpe ratio [3]: the probabilistic Sharpe ratio [2] against a benchmark equal to the expected maximum Sharpe ratio of N skill-less trials, computed with the best strategy's skewness and kurtosis and the cross-trial standard deviation of the 20 annualized Sharpe ratios. Its authors accept a strategy when DSR ≥ 0.95; we score $1-\text{DSR}$ as its p-value, so that rejection at 5% is exactly that rule.

Boot. An i.i.d. bootstrap [18] of the best strategy's demeaned returns with $B = 1{,}999$ resamples; the single-trial p-value $(h+1)/(B+1)$, where h counts resampled Sharpe ratios at or above the observed one, is Šidák-adjusted for 20 trials.

HAC-t. The best strategy's mean divided by a Newey–West (Bartlett kernel) standard error [40] with lag $\lfloor 4(T/100)^{2/9} \rfloor$, read against Student t with $T-1$ degrees of freedom and Šidák-adjusted.

SB-t. A stationary-bootstrap t test of the best strategy's mean [43] with mean block length $\text{round}(T^{1/3})$ and 4,999 resamples, each resample studentized by its own standard deviation; Šidák-adjusted.

The joint resampling tests use all 20 series and one stationary bootstrap of the whole search (the same resampled periods for every strategy, so the dependence between strategies is kept), with mean block length $b = \text{round}(T^{1/3}) = 8$ and $B = 1{,}000$ resamples. With $\bar{d}_k$ the mean of strategy k and $\bar{d}_k^*$ its mean in a resample:

RC. White's Reality Check [56]: $\max_k \bar{d}_k$ against the resampled distribution of $\max_k(\bar{d}_k^* - \bar{d}_k)$. It is not studentized.

SPA$_c$, SPA$_u$. Hansen's test for superior predictive ability [27] uses the statistic

$$T^{\text{SPA}} = \max\Big(0, \max_k \frac{\sqrt{T}\,\bar{d}_k}{\hat{\omega}_k}\Big), \qquad \hat{\omega}_k^2 = \hat{\gamma}_{0,k} + 2\sum_{i=1}^{T-1}\kappa_i\hat{\gamma}_{i,k},$$
$$\kappa_i = \Big(1-\frac{i}{T}\Big)\Big(1-\frac{1}{b}\Big)^i + \frac{i}{T}\Big(1-\frac{1}{b}\Big)^{T-i},$$

where $\hat{\omega}_k^2$ is the Politis–Romano estimate of the long-run variance at the bootstrap's block length and $\hat{\gamma}_{i,k}$ the lag-i sample autocovariance of strategy k (divisor T). As in Hansen's definition, the resampled statistic is $\max\big(0, \max_k \sqrt{T}(\bar{d}_k^* - g_k)/\hat{\omega}_k\big)$ with the *same* $\hat{\omega}_k$. The consistent p-value recentres at $g_k = \bar{d}_k$ when $\bar{d}_k \geq -\hat{\omega}_k\sqrt{2\ln\ln T/T}$ and at zero otherwise; the upper p-value recentres every strategy at its mean.

Step-SPA. The stepwise SPA test of Hsu et al. [33]: the step-down procedure of Romano and Wolf [45] applied to the same studentized statistics, with the same fixed $\hat{\omega}_k$ and consistent recentring. It returns a set of strategies rather than a p-value, so we score whether it declares at least one strategy superior at 5%. Romano and Wolf's own StepM differs: it re-estimates the standard errors in every resample [45, Algorithm 4.2].

In v1 every p-value of a joint test is the share of resamples whose statistic exceeds the observed one, $\#\{T^* > T\}/B$, and a test rejects when that share is at most 0.05; for an exact bootstrap this rule rejects with probability 5.09% rather than 5%, a difference of about half a Monte Carlo standard error. Finally, an *oracle* knows which strategy is skilled and applies a one-sided Newey–West t test to it alone. Its power is a natural benchmark: no best-of-N test in our tables exceeds it.

LET, HL, HL-Lo and DSR call the implementations behind the author's public calculator and API, and RC, SPA and Step-SPA the data-snooping implementation those use; the rest are written in the benchmark. No validator is tuned to the benchmark.

2.5 Performance measures, Monte Carlo error and runs

Each cell (family, skill level) of the main run, Null Zoo v1, has 10,000 searches. The Monte Carlo standard error of a rejection rate $\hat{p}$ from n searches is $\sqrt{\hat{p}(1-\hat{p})/n}$: 0.22 percentage points at 5% for n = 10,000. We flag a size as miscalibrated when it lies outside the 99% band around 5% (4.44–5.56% for 10,000 searches), so that a calibrated validator is flagged in about one size cell in a hundred.

The study consists of five *governed* runs, each of whose design, decision rules and predictions were committed to a public repository before it was run (the commits and times are listed in the data availability statement):

- **v1** (base seed 20261004). Its pre-registration fixed the families, the skill levels, the number of searches, a size bar (at most 6% at a nominal 5% in every family), a power bar (at least 45% at a Sharpe ratio of 2 under i.i.d. returns), the oracle and an ordered list of candidate tests, to choose default tests for the author's open-source validation software. The joint-test candidates were the two SPA p-values; the Reality Check was measured but not listed. For single series, the rule chose LET-Lo, the author's own statistic with Lo's factor, which missed the size bar only under negative skew (8.7% in v1 and 8.8% in v1b).

- **v1b** (base seed 20261005). v1 found the Reality Check close to its level where the listed SPA candidates were not. Adopting it on that evidence would have been a choice made after seeing the results, so a separate pre-registration added it to the list and repeated the run on fresh seeds; v1b's result decided the default whichever way it fell.

- **v2** (base seed 20261009). A second implementation of the families and the joint tests, in another language, written by the same author from the published definitions rather than translated from v1's code, tests why the studentized SPA test is oversized and where studentizing pays off. An unregistered pilot on another seed (four families, 2,000 searches each, disclosed with its results in the pre-registration) had already suggested the mechanism, so v2 confirms rather than discovers it. Its pre-registration fixed four experiments and 9 predictions. The only external implementation of a data-snooping test in the registered runs is arch 8.0.0 (experiment D), which does not studentize; arch's studentized test, after the change described in Section 4.5, was run only in an unregistered check.

- **v2b** (base seed 20261011). Registered after a review of the v2 draft, it maps the joint tests' size against the number of strategies and the block length, tests a block-studentized version of the bootstrap-t across the nine families, extends the unequal-volatility and poor-alternative designs, and adds the empirical illustration of Section 6. Small checks made during that review shaped it; they are disclosed with their results in its pre-registration, which fixed 7 predictions. From v2b on, every test whose statistic is floored at zero counts ties ($p = \#\{T^* \geq T\}/B$; Section 4.5).

- **v2c** (base seed 20261108). A confirmation run registered on 7 November 2026, after a review of this revision objected that its newest evidence was unregistered. It re-tests on fresh seeds, in both implementations, the unregistered checks listed below—arch's studentized SPA test after its correction, the author's corrected implementation, and the best-of-20 tests at small autocorrelations—adds the joint tests at small autocorrelations, and tests Proposition 2(b) against closed-form values committed with the pre-registration. Its pre-registration fixed 15 hypotheses with numeric bars, the seeds (checked by script to differ from every earlier run's), the scripts, and a rule for shortening the run that depends on time only (Section 5). Unlike v1–v2b, it was committed and signed but not pushed before the run, so its timing rests on the signed commit alone.

Besides the governed runs, the paper reports several *unregistered* checks, each with its script and result file in the repository and each run once: arch's code after the change merged on 6 October 2026 and after the one merged on 7 October 2026 (Section 4.5); a comparison of two ways of re-studentizing each resample; the author's data-snooping implementation before and after its correction, on 4,000 new searches per design; and the best-of-20 tests at small autocorrelations, 10,000 searches at each of five levels, with a measurement of the lag-one autocorrelation of the empirical illustration's rule returns (Section 3.3). They were not pre-registered; v2c repeats them under a pre-registration, and Section 5 reports both.

v1 and v1b use a xoshiro128** generator [7], v2 and v2b numpy's PCG64; every run seeds chunks of searches separately and merges them in a fixed order, so its output does not depend on the number of workers. An earlier published version (v0, 2,000 searches per cell with seven validators and eight families) and its 10,000-search robustness run are reported in the first version of this paper; v1 supersedes their size and power estimates. The paired bootstrap-resolution run and the dispersion diagnostics quoted in Section 3 come from those earlier, unregistered runs.

3 Results: single-series corrections

Table 1 reports the size of the single-series corrections in v1 and Table 2 raw and size-adjusted power at a Sharpe ratio of 2; the appendix gives the remaining validators, the other skill levels and the confirmation run v1b. Rates in the text are from v1 unless stated.

3.1 The baseline is calibrated, and the search itself costs power

Under independent normal returns the luck-equivalent-trials test is exact by construction: the t-statistic is Student t and the trials are independent. Its size is 4.9% (5.3% in v1b). The non-normal-error version (4.9%), the one-sided Harvey–Liu haircut (4.8%), the bootstrap (4.8%) and the Reality Check (5.1%) are calibrated too. The tests that estimate a long-run variance are slightly liberal even here (HAC-t 5.6%, the single-strategy oracle 5.6%, SB-t 6.1%): a kernel or block estimate of the variance is noisier than the sample variance, a theme Section 4 returns to.

Table 1: Size of the single-series corrections: percentage of skill-less searches (20 strategies, 504 daily returns) whose best strategy is called skilled at the nominal 5% level. Null Zoo v1, 10,000 searches per cell.

Family	LET	LET-Lo	LET-NN	HL	DSR	Boot	HAC-t	SB-t
IID normal	4.9	5.1	4.9	4.8	*0.1*	4.8	**5.6**	**6.1**
Student t_4	4.8	5.2	**6.7**	4.6	*0.1*	**6.3**	**5.7**	**7.9**
Skew −1.32	**8.3**	**8.7**	*4.4*	**8.1**	*0.1*	5.5	**9.0**	**6.7**
Skew +1.32	*3.0*	*3.4*	**6.1**	*2.9*	*0.1*	4.9	*3.7*	**6.6**
GARCH(1,1)	4.9	5.0	5.1	4.8	*0.1*	4.9	5.3	**6.0**
AR(1), $\phi = 0.2$	**19.5**	5.3	**19.4**	**19.1**	*0.3*	**18.4**	**7.4**	**7.6**
Two vol. regimes	5.0	5.2	5.5	4.9	*0.1*	5.4	5.1	**6.0**
Correlated, $\rho = 0.5$	*3.5*	*3.8*	*3.6*	*3.5*	*1.7*	*3.3*	*4.0*	*4.3*
Clusters, $\rho = 0.8$	*3.3*	*3.5*	*3.3*	*3.2*	*0.0*	*3.2*	*3.7*	*3.9*

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. HL is the Harvey–Liu haircut read one-sided (two-sided p halved); Boot uses 1,999 resamples and SB-t 4,999.

Selection, not the correction, costs most of the power. With a true Sharpe ratio of 2 over two years, the oracle that tests the skilled strategy alone rejects in 87.7% of searches; the best calibrated best-of-20 tests reach 52.9% (LET) and 53.5% (RC). The skilled strategy is the in-sample best in only 80.3% of these searches, so even a perfect correction would crown the wrong strategy in about one search in five; under AR(1) returns it is the best in 67.0%.

3.2 Autocorrelation breaks the tests that ignore it

With a lag-one autocorrelation of 0.2, the tests that treat returns as independent rejected about four times as often as their level: 19.5% for the luck-equivalent-trials test, 19.1% for the one-sided haircut, 19.4% with the non-normal standard error and 18.4% for the i.i.d. bootstrap. Positive autocorrelation makes the sample Sharpe ratio more variable than the i.i.d. formula assumes; the non-normal correction adjusts for the shape of the marginal distribution, which is normal here; and the i.i.d. bootstrap destroys the very dependence that inflates the sampling variance of the Sharpe ratio. The deflated Sharpe ratio, which also ignores autocorrelation, stays far below its level (0.3%) only because it rejects almost nothing anywhere.

Deflating the t-statistic by Lo's annualisation factor removes the excess here: the luck-equivalent-trials test falls to 5.3% (5.5% in v1b). But the AR(1) family is the factor's best case—it satisfies the assumed geometric decay exactly—and in the eight families without autocorrelation the factor raises the size by 0.10–0.47 points, because the estimated factor adds noise to the statistic. The remedies that make no parametric assumption about the dependence reduce the excess without removing it at this sample size: a Newey–West standard error (HAC-t, 7.4%) and a stationary-bootstrap t test (SB-t, 7.6%), and the joint resampling tests, whose blocks carry the dependence, are close but above the band (RC 6.3%, 6.4% in v1b).

3.3 How little autocorrelation it takes

A lag-one autocorrelation of 0.2 is a stress level. A further run, with v1's design and 10,000 skill-less searches at each of $\phi = 0$, 0.02, 0.05, 0.1 and 0.2, shows that much smaller values already break the best-of-20 tests that ignore autocorrelation (Table 3). At $\phi = 0.05$, where the i.i.d. standard error of the mean is only 5.1% too small, the one-sided haircut rejected 7.6% and the luck-equivalent-trials test 7.8%; at $\phi = 0.1$ they rejected 10.6% and 10.8%. Autocorrelations of this size are unremarkable.

Table 2: Power when one strategy has a true annualized Sharpe ratio of 2, Null Zoo v1 (percent). The oracle knows which strategy is skilled and tests it alone; its power is a natural benchmark, and no best-of-20 test in the table exceeds it.

Raw power

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC-t	RC	SPA$_c$
IID normal	87.7	52.9	52.9	9.2	51.2	54.0	53.5	56.9
Student t_4	88.7	54.0	54.0	10.3	53.3	55.5	52.0	58.2
Skew -1.32	87.5	55.3	55.4	7.7	46.4	56.1	56.1	61.7
Skew $+1.32$	89.3	52.6	53.1	11.8	58.2	53.6	51.4	54.3
GARCH(1,1)	87.7	54.7	54.8	11.3	53.2	55.6	51.5	58.4
AR(1), $\phi = 0.2$	77.3	60.3	35.4	8.0	58.4	40.3	38.5	42.5
Two vol. regimes	87.9	55.8	55.5	11.7	54.5	56.7	50.4	58.6
Correlated, $\rho = 0.5$	88.1	51.8	51.6	24.7	50.4	52.3	59.2	61.0
Clusters, $\rho = 0.8$	88.4	51.9	51.9	15.5	50.2	52.9	59.9	62.1

Size-adjusted power

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC-t	RC	SPA$_c$
IID normal	86.7	53.2	52.7	40.4	51.2	52.4	53.2	50.2
Student t_4	88.1	54.6	53.5	41.9	45.8	53.5	52.0	51.7
Skew -1.32	84.9	47.1	46.4	39.2	42.8	46.0	54.8	44.6
Skew $+1.32$	90.0	60.0	58.6	43.0	58.2	57.6	51.7	56.4
GARCH(1,1)	87.6	54.9	54.8	42.9	53.2	54.6	51.8	53.8
AR(1), $\phi = 0.2$	75.1	35.4	34.2	29.8	n/a	34.3	34.8	32.4
Two vol. regimes	87.4	55.6	54.9	43.5	51.3	56.1	50.8	56.5
Correlated, $\rho = 0.5$	87.5	56.8	55.8	40.8	54.9	56.1	56.9	54.2
Clusters, $\rho = 0.8$	88.1	58.9	58.1	51.3	57.0	58.1	58.7	55.8

Size-adjusted power is power at the cutoff at which the same test rejects at most 5% of the matching null cell; a user who does not know the return family cannot apply it, but it shows how well each statistic ranks skilled searches above skill-less ones. n/a: the test's smallest attainable p-value already occurs in more than 5% of the null cell, so no cutoff attains the level. The Monte Carlo standard error of raw power is at most 0.5 points. Size-adjusted power also carries the error of the estimated cutoff, and the joint tests' p-values lie on a grid of 1/1,000, so their achieved null rejection rate at the cutoff can be below 5%.

In the empirical illustration of Section 6, the daily returns of the 20 moving-average rules over buy-and-hold had a median lag-one sample autocorrelation of 0.075 across 1,010 rule-and-window pairs (interquartile range -0.004 to 0.157), at least 0.05 in 57% of them and in 57% of the windows' best rules (unregistered diagnostic); over the whole sample since 1927 it was 0.010–0.025 by rule and 0.046 for the market itself. Monthly managed-portfolio returns are often far more autocorrelated [22, 37]. An estimate of 0.05 from 504 daily returns, whose standard error is about $1/\sqrt{T} = 0.045$, is easily dismissed as noise, and the windowed estimates above are themselves that noisy.

The excess is predictable. Under an AR(1) the variance of $\sqrt{T}$ times a sample mean is inflated by $(1+\phi)/(1-\phi)$ in large samples—the factor behind Lo's [2002] correction of the Sharpe ratio and behind the "equivalent sample size" of tests of the mean [57]. Inserted into the best-of-N tail, it gives the size in closed form.

Proposition 2. *(a) Let N strategies have independent, stationary Gaussian AR(1) returns with mean zero and lag-one autocorrelation $\phi \in (-1, 1)$, and let a test reject when the largest t-statistic $\max_k \sqrt{T}\,\bar{x}_k/s_k$ exceeds a cutoff c chosen for i.i.d. returns (for the Šidák test, $\bar{F}_{T-1}(c) = 1-(1-\alpha)^{1/N}$; for the one-sided Bonferroni haircut, $\bar{F}_{T-1}(c) = \alpha/N$). Then each t-statistic converges in*

Table 3: How little autocorrelation it takes: size of the best-of-20 corrections that ignore autocorrelation, simulated and in closed form (Proposition 2), and with Lo's correction, when every strategy's daily returns follow a Gaussian AR(1) with lag-one autocorrelation ϕ (10,000 skill-less searches per level, 20 strategies, 504 days; percent at a nominal 5%).

		HL (one-sided)			LET		
ϕ	SE too small	Sim.	Closed	Lo	Sim.	Closed	Lo
0	0.0%	4.8	4.9	5.0	4.9	5.0	5.1
0.02	2.0%	**5.9**	5.8	5.4	**6.0**	5.9	5.5
0.05	5.1%	**7.6**	7.3	5.5	**7.8**	7.5	5.5
0.1	10.6%	**10.6**	10.5	5.0	**10.8**	10.7	5.1
0.2	22.5%	**19.0**	19.6	5.2	**19.3**	19.9	5.4

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. SE too small: the proportion by which the i.i.d. standard error of the mean understates the large-sample one, $\sqrt{(1+\phi)/(1-\phi)}-1$. Closed: equation (1). Lo: the same test after Lo's (2002) factor at the best strategy's estimated lag-one autocorrelation. HL is the Harvey–Liu haircut read one-sided, whose Bonferroni cutoff is slightly higher than LET's Šidák cutoff. The rows $\phi = 0$ and $\phi = 0.2$ repeat v1's i.i.d. normal and AR(1) cells on new seeds. Results at 1% and 10% are in the result file.

distribution to $N(0,\lambda)$ with $\lambda = (1+\phi)/(1-\phi)$, and approximating it by $\sqrt{\lambda}$ times a Student t_{T-1} variable gives the size

$$\pi(\phi) = 1 - \Big[1 - \bar{F}_{T-1}\Big(c\Big/\sqrt{(1+\phi)/(1-\phi)}\Big)\Big]^N, \tag{1}$$

which equals the i.i.d. size at $\phi = 0$ and increases with ϕ. At $\phi = 0$ its elasticity with respect to the standard-error ratio $\sqrt{\lambda}$ is

$$\frac{\mathrm{d}\ln\pi}{\mathrm{d}\ln\sqrt{\lambda}} = \frac{c\,f_{T-1}(c)}{\bar{F}_{T-1}(c)} \cdot \frac{N\bar{F}_{T-1}(c)\,[1-\bar{F}_{T-1}(c)]^{N-1}}{\pi(0)}, \tag{2}$$

which grows with the cutoff c and therefore with N.
(b) Suppose instead that $x_{k,t} = \sqrt{\rho}\,f_t + \sqrt{1-\rho}\,e_{k,t}$ with $0 \le \rho < 1$, where f and $e_1, \ldots, e_N$ are independent, stationary Gaussian AR(1) series with the same coefficient ϕ, mean zero and the same variance, so that every strategy is an AR(1) with coefficient ϕ and every pair of strategies has correlation ρ. Then the N t-statistics converge jointly to a Gaussian vector with variances λ and pairwise correlations ρ, and keeping the Student t_{T-1} marginal of part (a) gives the size

$$\pi(\phi,\rho) = 1 - \int_{-\infty}^{\infty} \Phi\Big(\frac{c' - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\Big)^N \varphi(z)\,\mathrm{d}z, \qquad c' = \bar{\Phi}^{-1}\Big[\bar{F}_{T-1}\Big(c\Big/\sqrt{(1+\phi)/(1-\phi)}\Big)\Big], \tag{3}$$

with Φ, $\bar{\Phi}$ and φ the standard normal distribution function, upper tail and density. It equals (1) at $\rho = 0$ and does not increase with ρ.

The proofs are in Appendix A. Equation (1) tracks the simulation closely: across the five levels it differs from the simulated size by at most 0.6 points for the haircut and 0.6 for the luck-equivalent-trials test, within 1.2 Monte Carlo standard errors for $\phi \le 0.1$; the largest gap is at $\phi = 0.2$, where the large-sample variance is least accurate at $T = 504$. Equation (2) explains why small values matter. At the 5% level a single test's cutoff is 1.65 and its elasticity 3.4; the best-of-20 cutoff is 2.81 and its elasticity 8.4 (11.2 for a hundred strategies). A standard error 1% too small thus raises the size of a best-of-20 test by about 8.4%, against 3.4% for a single test: at $\phi = 0.05$ a single test would reject 5.9% rather than 5%, the best-of-20 Šidák test 7.5%. None of this is deep—the

Table 4: Pre-registered confirmation (v2c): size of the best-of-20 single-series tests under AR(1) returns with lag-one autocorrelation ϕ and pairwise correlation ρ between strategies, simulated on fresh seeds (10,000 searches per row) and in closed form (percent at a nominal 5%).

	LET				HL (one-sided)			
ρ, ϕ	Sim.	(b)	(a)	Lo	Sim.	(b)	(a)	Lo
0, 0.02	**6.1**	5.9	5.9	5.6	**6.0**	5.8	5.8	5.5
0, 0.05	**7.7**	7.5	7.5	**5.6**	**7.5**	7.3	7.3	5.5
0, 0.1	**10.6**	10.7	10.7	5.2	**10.4**	10.5	10.5	5.1
0.3, 0	*4.0*	4.2	5.0	*4.2*	*4.0*	4.2	4.9	*4.1*
0.3, 0.05	**6.1**	6.1	7.5	4.7	**6.0**	6.0	7.3	4.6
0.3, 0.1	**8.3**	8.5	10.7	4.6	**8.2**	8.4	10.5	4.5
0.6, 0	*2.4*	2.8	5.0	*2.5*	*2.3*	2.8	4.9	*2.5*
0.6, 0.05	*3.7*	4.0	7.5	*2.8*	*3.6*	3.9	7.3	*2.7*
0.6, 0.1	**5.8**	5.5	10.7	*3.4*	**5.7**	5.4	10.5	*3.3*

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. (a): Proposition 2(a), equation (1), which assumes independent strategies; (b): Proposition 2(b), equation (3); the two coincide at $\rho = 0$. Both were computed and committed before the run. Lo: the same test after Lo's factor at the best strategy's estimated autocorrelation. Hypotheses H8–H10, H14 and H15.

variance inflation is Lo's, and the sensitivity of a far tail to its scale is elementary—but we have not found it stated for the multiple-testing corrections of strategy research, where the search pushes the cutoff into the far tail and so turns a negligible-looking autocorrelation into a large excess.

Part (a) assumes independent strategies, which real searches are not. Part (b) covers the simplest dependent case, a common factor: every strategy is an AR(1) with the same ϕ and every pair has the same correlation $\rho \geq 0$. Its assumptions are plain and strong—Gaussian returns, one ϕ for the factor and the idiosyncratic parts, one ρ for all pairs, the large-sample variance λ, and the dependence of the t-statistics approximated by that of their numerators. Two facts drive it. The autocorrelation inflates every strategy's variance by the same factor and leaves the correlation of the scaled means exactly ρ; and given the factor, the strategies are independent, so the size is part (a)'s expression averaged over the factor. The integral is classical for the maximum of equicorrelated normals [54]; what is new is only its use here. Correlation lowers the size (Slepian's inequality makes part (a) an upper bound), so correlated searches start conservative: at $\phi = 0$ and 5%, the Šidák test's size is 4.2% at $\rho = 0.3$ and 2.8% at $\rho = 0.6$. It does not lower the sensitivity to the standard error much: the elasticity (2), computed numerically for part (b), is 8.4 at $\rho = 0$, 7.8 at 0.3 and 7.4 at 0.6, because the cutoff stays in the far tail whatever the correlation. A pre-registered simulation confirmed part (b) at $\rho = 0.3$ and 0.6 (Section 5).

Lo's correction, applied at the best strategy's estimated autocorrelation, kept the size at 5.0–5.5% for the haircut and 5.1–5.5% for the luck-equivalent-trials test across all five levels, within the 99% band (4.44–5.56%) at every level, though at its upper edge for $\phi = 0.02$ and 0.05; in the pre-registered confirmation LET-Lo crossed that edge at $\phi = 0.05$ (5.61%; Section 5). These designs are the correction's best case: the AR(1) satisfies its geometric decay exactly, the strategies are independent, and the marginal distribution is normal.

3.4 Skew and fat tails

The Student-t tests are liberal under negative skew (8.3% for the luck-equivalent-trials test, 8.1% for the one-sided haircut, 9.0% for HAC-t) and conservative under positive skew (3.0%). With $T = 504$, the t-statistic of a skewed series is itself skewed: under negative skew a high sample mean

tends to come with a low sample variance, so the upper tail of t is heavier than Student t, and the reverse under positive skew.

The non-normal standard error of Mertens [38] moves the error rather than removing it: it is conservative under negative skew (4.4%) and liberal under positive skew (6.1%) and fat tails (6.7%). It was not designed to correct the null distribution: at a Sharpe ratio of zero its variance term equals one for every i.i.d. distribution, and skewness and kurtosis enter only because the term is evaluated at the estimate. Selection then makes that estimate misleading. Under t_4 returns, a symmetric distribution, the selected best series has a mean sample skewness of 0.58: a series tends to win because it had a large positive outlier. The single-strategy bootstrap tests inherit the same selection effect (Boot 6.3%, SB-t 7.9% under t_4): resampling a winner that owes its rank to an outlier reproduces the outlier's influence. No single-series validator is calibrated across negative skew, positive skew and fat tails; the Reality Check, which resamples all strategies jointly, is (5.4%, 4.9% and 4.9%).

Under GARCH volatility clustering and regime switching the Student-t tests stay near 5% (LET 4.9% and 5.0%): returns remain a martingale difference sequence with finite variance, and two years of daily data are enough for the t-test to be nearly exact.

3.5 Correlated strategies make the Bonferroni-type tests conservative

When every pair of strategies has correlation 0.5, or strategies come in clusters of five with correlation 0.8, the search contains fewer than 20 effectively independent trials, and every test that assumes 20 is conservative: 3.5% and 3.3% for the luck-equivalent-trials test, 3.3% and 3.2% for the bootstrap. The unused level costs power (LET 51.8% raw against 56.8% size-adjusted in the correlated family). The Reality Check, which resamples the strategies together and so sees their correlation, stays close to its level (5.6% and 5.5%; 5.5% and 5.7% in v1b).

3.6 The deflated Sharpe ratio at 0.95 tests a harder hypothesis

Bailey and López de Prado [3] accept a strategy when DSR ≥ 0.95. Scored this way, the deflated Sharpe ratio rejects at most 1.7% of skill-less searches (in the correlated family) and fails to accept 90.8% of i.i.d. normal searches that contain a strategy with a true Sharpe ratio of 2 over two years (75.3–92.3% across families; power 9.2% against 52.9% for the Šidák test).

The hurdle behind this conservatism, and how much of the shortfall comes from the threshold rather than the statistic, are worked out in Appendix B.1.

Two further results are in Appendix B: a bootstrap test's power depends on its number of resamples (with 400 resamples and a Šidák adjustment, power under i.i.d. normal returns was 46.7%, against 50.9% with 1,999), and the Harvey–Liu haircut read one-sided is the Šidák test up to the Bonferroni approximation.

4 Results: data-snooping tests

4.1 With block resampling, the fixed-variance SPA test is oversized

Hansen's studentized test for superior predictive ability, as Section 2 defines it—a Politis–Romano variance at the bootstrap's mean block length of 8, held fixed in the resamples—is oversized at two years of daily data and twenty strategies (Table 5). Its consistent p-value rejected more than the pre-registered 6% of skill-less searches at a nominal 5% in 7 of the 9 families in v1 and in 8 in v1b, and up to 12.6%. In v1 it rejected 7.5% under i.i.d. normal returns, 12.5% under negative skew

Table 5: Size of the joint resampling tests in Null Zoo v1 and in its confirmation run v1b (fresh seeds), 10,000 searches per cell each, 1,000 stationary-bootstrap resamples with mean block length 8.

Family	RC v1	RC v1b	SPA_c v1	SPA_c v1b	SPA_u v1	Step-SPA v1
IID normal	5.1	5.4	**7.5**	**7.8**	**7.4**	**7.5**
Student t_4	4.9	4.9	**7.3**	**7.6**	**7.2**	**7.3**
Skew −1.32	5.4	5.3	**12.5**	**12.6**	**12.2**	**12.5**
Skew +1.32	4.9	5.2	*4.3*	*4.3*	*4.2*	*4.3*
GARCH(1,1)	4.8	4.5	**6.5**	**6.3**	**6.4**	**6.5**
AR(1), $\phi = 0.2$	**6.3**	**6.4**	**8.9**	**9.1**	**8.8**	**8.9**
Two vol. regimes	4.8	5.1	**5.7**	**6.2**	5.5	**5.7**
Correlated, $\rho = 0.5$	**5.6**	5.5	**7.1**	**6.9**	**7.1**	**7.1**
Clusters, $\rho = 0.8$	5.5	**5.7**	**6.9**	**7.0**	**6.8**	**6.9**

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. Step-SPA is scored as a decision (at least one strategy declared superior at 5%).

Table 6: Why the studentized SPA test is oversized: size of variants of the consistent SPA test on the same searches (Null Zoo v2, experiment A; second implementation, 10,000 searches per cell, mean block length 8 unless marked 22).

Family	SPA_c	SPA_c-SD	SPA_c-BT	RC	SPA_c-U	SPA_c (22)	SPA_c-U (22)
IID normal	**7.5**	5.5	5.2	5.1	5.3	**12.9**	**5.8**
Student t_4	**6.9**	5.1	5.1	4.5	4.6	**11.9**	5.6
Skew −1.32	**12.3**	**10.2**	5.4	5.2	5.5	**17.8**	**5.7**
Skew +1.32	*4.2*	*2.8*	5.4	5.1	5.1	**8.6**	**6.0**
GARCH(1,1)	**6.6**	5.1	5.1	4.9	5.1	**11.3**	5.4
AR(1), $\phi = 0.2$	**9.4**	**6.9**	**6.6**	**6.1**	**6.2**	**13.5**	**6.3**
Two vol. regimes	**5.8**	4.8	5.0	4.5	4.8	**9.3**	5.0
Correlated, $\rho = 0.5$	**6.7**	**5.6**	5.5	5.4	5.4	**10.2**	**6.3**
Clusters, $\rho = 0.8$	**7.6**	**6.4**	**6.2**	**6.1**	**6.2**	**11.4**	**7.3**

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. SPA_c: Hansen's statistic studentized by a fixed Politis–Romano long-run variance. SPA_c-SD: studentized by the sample standard deviation instead. SPA_c-BT: every resample re-studentized by its own standard deviation (bootstrap-t). SPA_c-U: neither studentized nor floored at zero, the computation arch 8.0.0 performs. (22): mean block length 22, arch's default for 504 periods.

and 8.9% under autocorrelation. The upper p-value behaves almost identically, and the stepwise SPA test, whose first step uses the same statistic and recentring, matched the consistent SPA test to within 0.03 percentage points in every family of both runs. Romano and Wolf's StepM, which re-estimates the standard errors in every resample, was not run in v1; a stepwise test that does so, the author's corrected implementation, is scored in Section 4.5 (Table 9).

White's Reality Check, which uses the same bootstrap but does not studentize, rejected 4.5–5.7% of skill-less searches outside the AR(1) family in the two runs, and 6.3% and 6.4% under autocorrelation. Outside the AR(1) family it lies above the 99% band only in the correlated family in v1 and the cluster family in v1b. Unlike any single-series correction, it stays close to its level under negative skew, positive skew, fat tails, volatility clustering and correlated strategies at once.

This was not the comparison v1 was designed to make. Its pre-registration listed the two SPA p-values as candidates for the author's software and only measured the Reality Check, so the pre-registered rule chose the consistent SPA test, which failed the size bar in most families. Rather than switch to the Reality Check on that evidence, a second pre-registration (v1b) added it as the

Table 7: Size against the number of periods T and strategies k (Null Zoo v2, experiment B, 5,000 null searches per cell; mean block length $b = \text{round}(T^{1/3})$).

	IID normal				Skew −1.32			
Design	SPA_c	SPA_c-SD	SPA_c-BT	RC	SPA_c	SPA_c-SD	SPA_c-BT	RC
$T = 252$, $k = 20$ ($b = 6$)	**9.5**	**6.0**	5.5	5.2	**16.2**	**12.7**	5.8	5.0
$T = 504$, $k = 20$ ($b = 8$)	**7.1**	5.6	5.2	5.0	**12.1**	**10.0**	5.0	5.1
$T = 1{,}260$, $k = 20$ ($b = 11$)	**7.1**	**6.2**	**6.0**	**5.8**	**9.1**	**7.9**	5.1	5.1
$T = 2{,}520$, $k = 20$ ($b = 14$)	**6.2**	5.6	5.5	5.4	**7.7**	**6.9**	4.9	4.7
$T = 504$, $k = 5$ ($b = 8$)	**6.9**	**6.3**	**6.2**	**6.0**	**8.7**	**8.0**	5.4	5.7
$T = 504$, $k = 100$ ($b = 8$)	**9.5**	5.3	4.9	4.8	**18.4**	**12.3**	4.3	*3.9*

Bold: above the 99% Monte Carlo band around 5% (4.21–5.79 for 5,000 searches); italic: below it.

Table 8: arch after its fix (pull request 882, merge commit `bbfa879`): size of arch's studentized SPA test, which re-studentizes every resample by its own bootstrap-implied long-run variance, beside this paper's fixed-variance test and Reality Check on the same 2,000 new skill-less searches per family (unregistered check).

	Mean block length 8			arch's default, 22		
Family	arch_c	SPA_c	RC	arch_c	SPA_c	RC
IID normal	4.9	5.9	*3.5*	5.3	**10.4**	4.0
Student t_4	5.3	**7.1**	5.3	5.7	**11.9**	6.0
Skew −1.32	5.2	**12.3**	5.1	5.2	**17.8**	5.3
Skew +1.32	5.8	4.5	4.9	5.4	**8.5**	5.7
GARCH(1,1)	4.8	**6.6**	5.2	4.5	**10.8**	5.3
AR(1), $\phi = 0.2$	5.3	**9.2**	5.8	6.0	**13.1**	5.7
Two vol. regimes	4.0	4.9	4.0	3.8	**8.6**	4.7
Correlated, $\rho = 0.5$	5.0	**6.4**	5.4	5.9	**10.1**	6.1
Clusters, $\rho = 0.8$	5.9	**7.2**	**6.5**	**6.4**	**10.5**	**7.3**

Bold: above the 99% Monte Carlo band around 5% (3.74–6.26 for 2,000 searches); italic: below it. arch_c: arch 8.0.1.dev102+gbbfa879a9 with its defaults (`studentize=True`, `nested=False`, stationary bootstrap, 1,000 resamples), consistent p-value. SPA_c: Hansen's statistic with the Politis–Romano variance held fixed, the computation arch adopted in pull request 871 and replaced in pull request 882. Both are unreleased; the latest release, 8.0.0, does not studentize (Table 21).

first candidate and repeated the run on fresh seeds. v1b confirmed the pattern; its rule chose the Reality Check, which missed the bar only in the AR(1) family. In v2, the second implementation reproduced v1b's sizes of the consistent SPA test and the Reality Check in all nine families to within three Monte Carlo standard errors (prediction P1 held; the largest difference was 0.62 points).

4.2 Why: a variance estimate the bootstrap holds fixed

The studentized statistic divides each mean by $\hat{\omega}_k$, estimated once from the sample, and the resampled statistics divide by the same $\hat{\omega}_k$. The bootstrap therefore reproduces the sampling variation of the means but not that of the estimate in the denominator. Romano and Wolf [45, footnote 22] note that this choice is only first-order consistent and that it is preferable to compute the standard errors from the bootstrap data [25], as their StepM does; Hall and Wilson [26] state the general guideline. What follows measures the size cost of not doing so at the sample lengths, block lengths and numbers of strategies of strategy research.

The size cost has a simple form under i.i.d. normal returns. For one series $x_1, \ldots, x_T$ with

Table 9: The author's implementation before and after the fix: size of the consistent SPA test, the stepwise SPA test and the Reality Check on the same 4,000 new skill-less searches per design and the same bootstrap draws (unregistered check; 1,000 resamples, mean block length 8).

	Before (v1 code)		After (bootstrap-t, ties)		
Family	SPA_c	Step	SPA_c	Step	RC
IID normal	**7.2**	**7.2**	5.3	5.3	5.2
Student t_4	**7.2**	**7.2**	5.3	5.3	5.2
Skew -1.32	**12.0**	**12.0**	5.2	5.2	5.1
Skew $+1.32$	4.3	4.3	5.4	5.4	4.9
GARCH(1,1)	**6.6**	**6.6**	5.4	5.4	4.8
AR(1), $\phi = 0.2$	**10.0**	**10.0**	**6.9**	**6.9**	**6.7**
Two vol. regimes	5.9	5.8	5.0	5.0	4.7
Correlated, $\rho = 0.5$	**7.7**	**7.6**	**6.0**	**6.0**	**6.0**
Clusters, $\rho = 0.8$	**7.5**	**7.5**	**6.2**	**6.2**	**6.1**
Poor alternatives	**7.3**	5.1	5.0	5.0	*0.3*

Bold: above the 99% Monte Carlo band around 5% (4.11–5.89 for 4,000 searches); italic: below it. Before: the code that scored v1 and v1b (Politis–Romano standard deviation held fixed, $p = \#\{T^* > T\}/B$). After: canlicapital pull request 562, each resample divided by its own standard deviation, ties counted and the stepwise critical value floored at zero. Step: at least one strategy declared superior at 5%. RC is unchanged by the fix. Poor alternatives: one strategy with mean zero and 19 with a true annualized Sharpe ratio of -3, i.i.d. normal returns.

sample mean $\bar{x}$, let $\hat{\gamma}_i = T^{-1}\sum_{t=1}^{T-i}(x_t - \bar{x})(x_{t+i} - \bar{x})$ and let $\hat{\omega}^2 = \hat{\gamma}_0 + 2\sum_{i=1}^{T-1}\kappa_i\hat{\gamma}_i$ be the Politis–Romano estimate, with weights $\kappa_i = (1 - i/T)(1-p)^i + (i/T)(1-p)^{T-i}$ and $p = 1/q$ for mean block length q. Define

$$r_T(q) = \frac{T-1}{T} - \frac{2}{T^2}\sum_{i=1}^{T-1}\kappa_i\,(T-i), \qquad v_T(q) = \frac{2}{T}\Big(1 + \frac{2}{T}\sum_{i=1}^{T-1}\kappa_i^2\,(T-i)\Big). \tag{4}$$

Lemma 1. *(a) If $x_1, \ldots, x_T$ are i.i.d. with variance σ^2, then $E[\hat{\omega}^2] = \sigma^2 r_T(q)$. (b) If they are also normal, $\bar{x}$ is independent of $\hat{\omega}^2$, and $\mathrm{Var}(\hat{\omega}^2) = \sigma^4 v_T(q) + O(T^{-2})$. (c) Both $1 - r_T(q)$ and $v_T(q)$ increase with q.*

Proposition 1. *Let N strategies have independent i.i.d. normal excess returns with mean zero, and let the fixed-variance test reject when $\max_k \sqrt{T}\,\bar{x}_k/\hat{\omega}_k$ exceeds $z^* = \Phi^{-1}\big((1-\alpha)^{1/N}\big)$, the first-order limit of its bootstrap critical value. If $\hat{\omega}_k^2/(\sigma_k^2 r_T)$ is approximated by χ^2_ν/ν with $\nu = 2/v_T$, its rejection probability is*

$$\pi_T(q) = 1 - \Big[1 - \bar{F}_\nu\big(z^*\sqrt{r_T(q)}\big)\Big]^N, \tag{5}$$

where $\bar{F}_\nu$ is the upper tail of Student's t with ν degrees of freedom. Then $\pi_T(q) > \alpha$ for every q and T, $\pi_T(q)$ increases with q, and $\pi_T(q) \to \alpha$ as $T \to \infty$ with q fixed.

The proofs are in Appendix A. The bootstrap cannot correct either term. Under the stationary bootstrap the conditional variance of $\sqrt{T}$ times a resampled mean is the Politis–Romano estimate itself [43, Lemma 1], so a resample divided by the same $\hat{\omega}_k$ has unit variance whatever the error in $\hat{\omega}_k$, and its critical value stays near z^*. At $T = 504$, $\hat{\omega}$ has a coefficient of variation of 8.6% at block length 8 and 14.3% at 22, against 3.1% for the sample standard deviation, so $\nu \approx 68$ and 24; $\hat{\omega}^2$ is biased downward by 2.9% and 8.2%; and with $z^* = 2.80$ for 20 strategies, (5) gives 7.2% and 12.1%, against 7.5% and 12.9% measured in experiment A: 89% and 91% of the excess. Without the bias term ($r_T = 1$) it would give only 6.5% and 9.4%. The formula ignores the variability of the bootstrap's own critical value, which Appendix B.4 measures. With block length 1, the setting of Hansen's own simulations, the Politis–Romano variance is the sample variance, and (5) gives

Table 10: Pre-registered confirmation (v2c): size of the joint tests on fresh seeds, nine families and three small autocorrelations (percent at a nominal 5%). Left: 3,000 searches per row, arch at the merge commit of pull request 882 and this paper's fixed-variance SPA test and Reality Check on the same searches. Right: 3,000 other searches per row, the author's implementation as v1 ran it and as corrected, with the same bootstrap draws.

	arch 882		Fixed-variance SPA_c		RC	v1 code	Corrected		
Family	8	22	8	22	8	SPA_c	SPA_c	StepM	RC
IID normal	5.4	**6.6**	**7.7**	**13.0**	5.3	**7.1**	4.9	4.9	4.7
Student t_4	4.5	4.9	5.8	**11.2**	4.0	**6.2**	4.8	4.8	4.3
Skew −1.32	5.8	5.8	**12.4**	**18.2**	5.0	**12.9**	**6.1**	**6.1**	5.8
Skew +1.32	4.9	5.4	4.0	**7.7**	5.0	*3.2*	4.7	4.7	4.3
GARCH(1,1)	5.4	5.3	**6.8**	**11.1**	4.9	**6.9**	5.7	5.7	5.1
AR(1), $\phi = 0.2$	**6.1**	**6.3**	**9.8**	**14.1**	**6.4**	**8.9**	**6.3**	**6.3**	**6.0**
Two vol. regimes	4.3	*3.7*	5.5	**8.7**	4.5	5.7	4.9	4.9	4.8
Correlated, $\rho = 0.5$	5.0	**6.1**	**6.7**	**10.3**	5.5	**6.9**	5.7	5.7	5.7
Clusters, $\rho = 0.8$	6.0	**6.7**	**7.8**	**11.4**	5.6	**6.4**	5.0	5.0	5.0
AR(1), $\phi = 0.02$	5.9	5.9	**7.7**	**13.1**	5.8	**8.0**	5.8	5.8	5.6
AR(1), $\phi = 0.05$	5.4	6.0	**7.9**	**12.8**	5.6	**7.9**	5.9	5.9	5.4
AR(1), $\phi = 0.1$	4.8	5.4	**8.3**	**12.3**	5.5	**8.1**	**6.3**	**6.3**	5.8

Bold: above the 99% Monte Carlo band around 5% (3.98–6.02 for 3,000 searches); italic: below it. 8, 22: mean block length (22 is arch's default for 504 periods); the author's implementation uses 8. arch: consistent p-value, every resample studentized by its own long-run variance. StepM: at least one strategy declared superior at 5%. Pre-registration commit `c0996a70714f`; hypotheses H1–H7 and H11–H13 are scored on this table (Table 28).

almost no excess (5.2%). A bootstrap-t, which re-estimates the standard error in every resample, reproduces the variability of the denominator and removes the first-order term [25].

Experiment A tests the account directly (Table 6). Studentizing by the sample standard deviation instead, with the bootstrap unchanged, removes most of the excess in the families with symmetric, serially uncorrelated and independent strategies: 5.5% under i.i.d. normal returns, 5.1% under t_4, 5.1% under GARCH and 4.8% with volatility regimes. With correlated strategies it removes part of the excess (6.7% to 5.6%, and 7.6% to 6.4% in the cluster family). Under negative skew it removes little (10.2%), because a second effect appears: a high sample mean tends to come with a low sample standard deviation, so the studentized statistic has a heavier upper tail than any fixed denominator can reproduce. A bootstrap-t that re-studentizes every resample by its own standard deviation reproduces both effects, in line with Hall and Wilson's guideline. Under negative skew its size was 5.4%, and outside the AR(1) family 5.0–6.2%, about the Reality Check's (4.5–6.1%) rather than calibrated in every family: both are above the band in the cluster family (prediction P3 held, P4 held). With the long block that arch uses by default ($\lfloor\sqrt{T}\rfloor = 22$), the fixed-variance test rejected 12.9% of i.i.d. normal searches and 17.8% under negative skew (P6 held).

The mechanism was suggested by an unregistered pilot (four families, 2,000 searches each, disclosed with its results in the v2 pre-registration) and confirmed in the registered run; v2 is a confirmation on fresh seeds, not a test of a hypothesis formed before any data were seen.

4.3 How the excess depends on the sample and the search

The excess of the fixed-variance test is a finite-sample effect: it shrinks as the sample grows and grows with the number of strategies (Table 7; prediction P7 held). Under i.i.d. normal returns it rejected 9.5% of skill-less searches with one year of daily data, 7.1% with two and 6.2% with ten; with two years of data it rejected 6.9% of searches of five strategies and 9.5% of searches of a

hundred. Under negative skew the same figures were 16.2%, 12.1%, 7.7%, 8.7% and 18.4%. The bootstrap-t and the Reality Check stayed closer to 5% (4.3–6.2% and 3.9–6.0% over the 12 cells) but not within the band everywhere: the bootstrap-t is above it in 2 cells and the Reality Check outside it in 3.

Earlier simulations did not report this excess, for identifiable reasons. Hansen [27] used $q = 1$, i.i.d. resampling, "in accordance with the lack of time-dependence" in his designs (p. 22 of the working-paper version); the Politis–Romano variance is then the sample variance, which is our SPA_c-SD column, close to nominal under i.i.d. returns. The excess appears only with block resampling, which dependent returns require and which arch uses by default. The working-paper version of Hsu et al. [33] reports family-wise error rates of the studentized stepwise SPA test of 3.2%, 1.8% and 1.2% with 90, 900 and 9,000 neutral strategies and 1,000 periods, from 500 replications (a standard error of about one point) with 500 resamples; their weight function, as printed, gives the j-th autocovariance the weight $(1 - Q)^j$ with $Q = 0.9$, which, if computed that way, makes their studentizer almost the sample standard deviation. Romano and Wolf [45] found satisfactory control of the family-wise error for their re-studentized StepM with i.i.d. data, 100 periods, 40 strategies and Efron's bootstrap. We have not reproduced these designs. The general problem—tests studentized by a noisy long-run variance are oversized in finite samples—is well known in heteroskedasticity- and autocorrelation-robust inference [34, 35, 53], including tests of equal predictive accuracy [15], and Gonçalves and Vogelsang [23] show that a block bootstrap which recomputes the long-run variance in each resample with the formula used on the data reproduces the fixed-b approximation and can be much more accurate than the normal one. Distinct from this finite-sample effect, Frederiksen et al. [20] show that the Diebold–Mariano test can be severely oversized even asymptotically when loss differentials have infinite variance.

4.4 The bootstrap's own variance and long blocks

Every block-bootstrap test, studentized or not, inherits the noisy, downward-biased Politis–Romano estimate through its critical value. Experiment E (Appendix B.4) shows the effect is small at block length 8 but makes the Reality Check and the bootstrap-t liberal at block length 22 with up to 20 strategies (6.4–6.6% and 6.3–6.7%), while the fixed-variance test compounds the two effects (7.5% with twenty strategies at block length 8). Re-studentizing each resample by its natural block variance (experiment F, Appendix B.5) cures the bootstrap-t's excess under autocorrelation but is conservative elsewhere.

4.5 What this means for software

The latest release of the open-source Python package arch [48] does not studentize its SPA statistic. In version 8.0.0, `SPA.compute()` and the routine that simulates the resampled statistics never read the `studentize` attribute; the option only sets a label in the test's information, and the variance estimate enters only the consistent recentring threshold. Experiment D confirms it: on 12,000 search-and-block-length pairs, arch's p-values were identical with `studentize=True` and `studentize=False` on every pair, and agreed with this paper's unstudentized consistent test to within Monte Carlo error (mean absolute difference 0.014–0.015, against 0.014 expected from two independent bootstraps of 1,000 resamples; Table 21). These statements concern `SPA` with `nested=False`. Arch's SPA test is thus the Reality Check with Hansen's consistent recentring. It is much less oversized than the fixed-variance test, but not calibrated at its default block length of 22: our implementation of the same computation rejected 5.0–7.3% of skill-less searches outside the AR(1) family at block length 22 (4.6–6.2% at block length 8), and arch itself rejected 6.5% of

i.i.d. normal searches at its default in experiment D. Prediction P5, which held, was registered and scored at block length 8 only.

A pull request opened on 27 September 2026 and merged on 6 October 2026 divides both the statistic and the resampled values by the fixed standard error $\sqrt{\hat{\omega}^2/T}$, in SPA and in StepM, whenever `studentize` is true, which is the default [55]. At arch's default block length and apart from the floor at zero, that is the computation measured above at 12.9% and 17.8%. In an unregistered check we ran the merged code itself (arch 8.0.1.dev83+g818d64a1b, merge commit 818d64a) on 2,000 new null searches per family. At its default block length it rejected 13.5% of i.i.d. normal, 19.1% of negatively skewed and 13.3% of AR(1) searches, against 13.1%, 19.1% and 13.5% for our implementation of the fixed-variance test on the same searches; at block length 8 the figures were 8.3%, 13.4% and 9.3%. Released arch 8.0.0, on the searches of experiment D, rejected 6.5%, 5.5% and 7.4%.

We reported these figures to the maintainers before the change was released [10] and proposed a bootstrap-t that re-studentizes every resample by its own standard deviation [11]. The maintainer replied that a nested bootstrap would be too expensive and suggested a closed-form estimator within each resample. Our unregistered comparison of the two per-resample studentizers, run that day on 2,000 new null searches per family, found both close to the level where the fixed-variance test was not: at block length 8, the resample's standard deviation gave 5.4%, 5.5% and 6.4% (i.i.d. normal, negative skew, AR(1)) and the resample's own Politis–Romano variance 6.1%, 6.0% and 5.7%, against 8.3%, 12.8% and 9.2% for the fixed variance; at block length 22 the three gave 6.0–6.1%, 5.5–6.5% and 12.7–19.4% across the three families. The same day the maintainer merged his own change [47]: with `studentize=True`, arch now divides every resample by the square root of the long-run variance implied by the bootstrap, computed from that resample—for the stationary bootstrap, the Politis–Romano estimate with the weights κ_i of Section 2—so the observed and resampled statistics are studentized alike, as Romano and Wolf recommend. The change is scheduled for arch 9.0; our pull request was closed as superseded, and the issue as fixed.

We then ran arch's merged code (8.0.1.dev102+gbbfa879a9, merge commit bbfa879), with all its defaults, on 2,000 new null searches in each of the nine families (Table 8; unregistered). It is the first external implementation of a studentized data-snooping test run in this study. At its default block length of 22 it rejected 3.8–6.4% of skill-less searches (5.3% i.i.d. normal, 5.2% negative skew, 6.0% AR(1)), against 8.5–17.8% for the fixed-variance test and 4.0–7.3% for the Reality Check on the same searches; at block length 8, 4.0–5.9%, against 4.5–12.3% and 3.5–6.5%. The fix removes the excess the paper attributes to the fixed variance, in code we did not write. Under autocorrelation, where the standard-deviation bootstrap-t was liberal, arch's per-resample long-run variance rejected 5.3% at block length 8 and 6.0% at 22, in line with the comparison above; it is the block studentization that treats the observed and resampled statistics alike which Appendix B.5 calls for, here tested only in this unregistered check. Its size was above the band only in the cluster family at block length 22, where the Reality Check was too. The registered evidence for re-studentizing remains experiments A, B, E and F; these checks show only that two independently written production implementations behave as the registered computation does. The confirmation run v2c then repeated both on fresh seeds under a pre-registration, and every hypothesis about them held (Section 5).

The same fix, applied to the implementation that scored v1 and v1b, shows the size of the change for a user. Its corrected version [12] re-studentizes every resample by its own standard deviation, counts ties and floors the stepwise critical value at zero. On 4,000 new skill-less searches per design, with the same bootstrap draws for both versions (Table 9; unregistered), the consistent SPA test's size went from 7.2% to 5.3% under i.i.d. normal returns, from 12.0% to 5.2% under negative skew and from 10.0% to 6.9% under autocorrelation, where the Reality Check rejected

6.7%; outside the AR(1) family it went from 4.3–12.0% to 5.0–6.2%. The fix changed two things at once, but in the nine families, where every strategy's mean is zero, a search with no strategy above the consistent threshold is too rare for counting ties to matter (see below), so the change there is the studentization's. The stepwise test, now a studentized StepM in the sense of Romano and Wolf [45] with Hansen's consistent recentring, followed the SPA test to within 0.03 points in every design. In the poor-alternative design, where ties matter, the SPA test went from 7.3% to 5.0%. The corrected test is not calibrated everywhere: like the Reality Check, it is above the band in the AR(1), correlated and cluster families, where its size was within 0.6 points of the Reality Check's in every family.

A second, smaller issue concerns ties. With the statistic floored at zero, $T^{\mathrm{SPA}} = 0$ whenever no strategy beats the benchmark in sample. Strategies below the consistent threshold are recentred at zero rather than at their means, so if every strategy falls below it, the floored resampled statistics are mostly zero as well, and a p-value with a strict inequality, $\#\{T^* > T\}/B$—the definition in Hansen [27] (p. 20 of the working-paper version)—counts only the few positive resamples: the test rejects although no strategy did better than the benchmark. With a single strategy whose mean is exactly zero this happens with probability $\Phi(-\sqrt{2\ln\ln T}) = 2.8\%$, and in experiment E the strict-inequality version of the fixed-variance test had size 9.0% with one strategy at block length 8, against 5.5% with ties counted. In v2 the defect caused the second failed prediction: P9 expected the bootstrap-t version to keep its size in a design whose other strategies were clearly poor, and it rejected 7.3% (Table 20). An exploratory check after that run, on a fresh seed, traced the excess to the ties (searches with $T = 0$ contributed 2.5 points of the bootstrap-t test's 7.8%), and v2b registered the fix. Counting ties, $p = \#\{T^* \geq T\}/B$, which gives $p = 1$ whenever $T = 0$, the floored tests' size in that design was 5.0% and 4.9%, against 7.4% with the strict inequality (P16 held). The implementation behind v1, the author's, also used the strict inequality; its corrected version counts ties (see above). Under v1's null hypotheses, where every strategy has mean zero, the case is too rare to matter (at most 1.7 searches in 100,000 under a normal approximation), but it matters for single-strategy calls and for searches with many poor variants.

4.6 Where studentizing and recentring pay off

Studentizing pays when strategies' volatilities differ and the skilled strategy is among the less volatile, because it ranks by t-statistic; consistent recentring pays when the other strategies are clearly poor, provided ties are counted. Experiment C2 quantifies both (Appendix B.6).

4.7 Power

Under i.i.d. normal and correlated returns, studentizing buys no power: at a Sharpe ratio of 2 the bootstrap-t version and the Reality Check had similar size-adjusted power (53.4% and 53.2% under i.i.d. normal returns, 56.5% and 56.5% in the correlated family; Table 19). Where realized volatilities differ across strategies even though their unconditional variances are equal, the bootstrap-t gains a few points (54.3% against 50.7% under GARCH, 54.8% against 50.4% with volatility regimes, 54.5% against 52.7% under t_4). Skew pulls the two apart in opposite directions: under negative skew the Reality Check was ahead (54.9% against 46.7%), under positive skew the bootstrap-t (57.8% against 49.1%). The fixed-variance test's apparent power advantage (raw 56.9% in v1, against 53.5% for the Reality Check) disappears once its excess size is removed (50.2% against 53.2%).

4.8 The pre-registered predictions

Of the 9 predictions registered before the v2 run, 7 held; of the 7 registered before v2b, 6 held (Tables 26 and 27). The three that failed are reported above: P2 and P9 in v2, and P12 in v2b. The explanation of P9 came from an exploratory check made after the v2 run; v2b then tested it as a registered prediction (P16).

5 A pre-registered confirmation

The newest evidence in Sections 3.3 and 4.5—arch's studentized SPA test after its correction, the author's corrected implementation, and the best-of-20 tests at small autocorrelations—came from checks run once without pre-registration, and the joint tests had been run at only one autocorrelation, 0.2. Run v2c repeats those checks on fresh seeds under a pre-registration, extends the joint tests to small autocorrelations, and tests Proposition 2(b). The pre-registration (signed commit `c0996a70714f`, 2026-11-07T18:24:34Z) fixed 15 hypotheses with numeric bars, the scripts that run and score them, the seeds and the closed-form predictions, before any search was drawn. It has three parts, all with 20 strategies, 504 days and a nominal 5% level:

S, the nine families. On 3,000 new null searches per family, arch at the merge commit of pull request 882 with its defaults, at block lengths 8 and 22, beside this paper's fixed-variance SPA test and Reality Check on the same searches (Python); on 3,000 other searches per family, the author's implementation as v1 ran it and as corrected, with the same bootstrap draws (JavaScript).

L, small autocorrelations. Independent Gaussian AR(1) strategies with $\phi = 0.02$, 0.05 and 0.1: the same joint tests on the same numbers of searches, and the four single-series tests of Table 3 on 10,000 searches per level.

C, correlated strategies. Strategies built as in Proposition 2(b) with $\rho = 0.3$ and 0.6 and $\phi = 0$, 0.05 and 0.1, 10,000 searches per cell, scored against equation (3).

Both parts finished within the registered time limit (33.9 and 7.1 minutes with twelve workers), so the shortening rule did not bind. Between the commit and the run both scripts were smoke-tested on another seed, as the pre-registration allows; no code changed. Because every seed is fixed in the committed pre-registration and the run is deterministic, a re-run reproduces the result files—the JavaScript part, run again with 10 workers instead of 12, gave an identical result file apart from its run time—so a different outcome could not have been obtained by running again without changing the committed seeds. Of the 15 hypotheses, 14 held and one, H10, did not (Table 28).

Re-studentizing, confirmed. Across the nine families arch's corrected test rejected 4.3–6.1% of null searches at block length 8 (H1) and 3.7–6.7% at its default of 22 (H2), against 4.0–12.4% and 7.7–18.2% for the fixed-variance test on the same searches; at block length 22 the fixed-variance test rejected at least 2.3 points more in every family (H3; Table 10). Under negative skew, the family with the largest excess, the fixed-variance test rejected 12.4% and arch 5.8% at block length 8, and the author's implementation went from 12.9% to 6.1% (H6). The corrected implementation rejected 4.7–6.1% outside the AR(1) family and 6.3% within it, never more than 0.6 points from the Reality Check on the same searches (H4, H5), and its StepM rejected at the same rate as its SPA test in every design (H7). Not every corrected test sat inside the 99% band (3.98–6.02% for 3,000 searches): arch at block length 22 was above it in the i.i.d. normal, AR(1), correlated and

cluster families, the Reality Check at block length 22 in the AR(1), correlated and cluster families, and the corrected implementation in the negative-skew and AR(1) families. These residual excesses are small next to the fixed-variance test's and of the size the unstudentized Reality Check shows with long blocks and correlated strategies (Section 4); the corrected tests are close to their level, not exact.

Joint tests at small autocorrelations. At $\phi = 0.02$–0.1 the tests that resample whole blocks and re-studentize, or do not studentize, stayed close to their level, while the single-series tests did not (H11–H13): arch rejected 4.8–5.9% at block length 8 and 5.4–6.0% at 22, the Reality Check 5.5–5.8% and the corrected SPA test and StepM 5.8–6.3%, against 7.55% and 10.42% for the one-sided haircut at $\phi = 0.05$ and 0.1. The fixed-variance test was oversized at every level (7.7–8.3% at block length 8, 12.3–13.1% at 22; v1's implementation 7.9–8.1%). A block bootstrap carries a small autocorrelation into its resamples; a t-statistic read against Student's t does not.

Proposition 2 and Lo's factor. The single-series sizes confirmed the earlier run: LET rejected 6.09%, 7.70% and 10.61% at $\phi = 0.02$, 0.05 and 0.1 (H8), within 1.0 Monte Carlo standard errors of Proposition 2(a) for both tests (H9; Table 4). H10 failed. It required Lo's factor to keep both tests inside the 99% band at every level; the haircut version did (5.10–5.51%), but LET-Lo rejected 5.61% at $\phi = 0.05$, just above the band's upper edge, and 5.56% at $\phi = 0.02$. With the earlier run, which put it at the edge at the same two levels, the evidence is that Lo's factor at an estimated autocorrelation leaves LET about half a point liberal at small ϕ—it removes most of the excess, not all of it.

Correlated strategies. Equation (3) matched the simulated sizes within 2.7 standard errors in all six cells (H14), and the independence formula overstated them at $\rho = 0.6$ by at least 16.8 standard errors (H15). At $\rho = 0.6$ the agreement has a pattern: the closed form was 2.5–2.7 standard errors above the simulation at $\phi = 0$ and 1.3–1.5 below it at $\phi = 0.1$, so the simulated size rose somewhat faster with ϕ than the Gaussian-copula approximation predicts. Correlation makes the Bonferroni-type tests conservative, but it does not protect them: at $\rho = 0.6$, LET rejected 2.4% without autocorrelation and 5.8% at $\phi = 0.1$, and at $\rho = 0.3$ it rose from 4.0% to 8.3%. A small autocorrelation first spends the margin that correlation gives, then exceeds the level.

6 An empirical illustration

The simulations say which tests keep their level; they do not say whether the choice among them changes a conclusion on real data. As a small illustration, registered with v2b but without a prediction, we apply the tests to a classic search: 20 moving-average timing rules on the US stock market [9, 52]. The data are the daily market and risk-free returns in the Fama/French 3 Factors file of the Kenneth R. French Data Library [19, 21], from 4 March 1927 to 31 August 2026; the file's SHA-256 is recorded with the results. Rule L, for $L = 10, 20, \ldots, 200$ trading days, holds the market on day $t + 1$ when a total-return index of the market closes above its L-day moving average on day t, and the risk-free asset otherwise. The series tested is each rule's return minus the market's, so the null hypothesis is that no rule beats buy-and-hold. We split the sample into 51 non-overlapping windows of 504 trading days, the length of the simulated searches, from the first day on which every rule is defined. Each window is tested with the joint tests at mean block length 8 and 1,000 resamples, with the fixed-variance and unstudentized tests also at block length 22, and with LET, the Šidák-adjusted t test on the best rule's Sharpe ratio.

Table 11: Twenty moving-average timing rules against buy-and-hold on the US market, in the 7 of 51 non-overlapping two-year windows in which at least one test rejects at 5% (Null Zoo v2b, part G; p-values).

	Mean block length 8					22		
Window	RC	SPA$_c$-U	SPA$_c$	SPA$_c$-BT	SPA$_c$-NB	SPA$_c$-U	SPA$_c$	LET
Jul 1930–Apr 1932	**0.038**	**0.038**	**0.020**	**0.030**	**0.027**	**0.015**	**0.008**	0.148
Dec 1958–Dec 1960	0.084	0.084	0.115	0.097	0.107	**0.032**	0.056	0.572
Dec 1960–Dec 1962	0.072	0.072	**0.047**	**0.050**	**0.024**	0.101	0.084	0.486
Jan 1969–Jan 1971	**0.019**	**0.019**	**0.010**	**0.007**	**0.003**	**0.022**	**0.016**	**0.008**
Jan 1971–Jan 1973	0.063	0.063	**0.049**	0.115	0.094	**0.031**	**0.032**	0.190
Jan 1973–Jan 1975	**0.010**	**0.010**	**0.000**	**0.004**	**0.003**	**0.015**	**0.001**	**0.022**
Jan 1977–Jan 1979	0.078	0.078	0.074	0.068	0.055	**0.042**	**0.048**	0.248
Windows rejected	3	3	5	4	4	6	5	2
Full sample†	**0.050**	**0.050**	**0.042**	**0.038**	**0.031**	**0.043**	**0.039**	0.316

p-values with 1,000 stationary-bootstrap resamples; bold: at most 0.05. LET is the Šidák-adjusted t test on the best rule's Sharpe ratio, with no resampling. Floored tests count ties. A superscript gives the number of rules used when some rules never left the market in a window and so had no excess return. Windows rejected: of all 51 windows. †4 March 1927 to 31 August 2026, 26,117 days, mean block length 30 for the first five tests. Every window is in Table 25.

The tests agree on most windows: none rejects at 5% in 44 of the 51, and all reject in 2 (Table 11; every window is in Table 25). In the other 5 the verdict depends on the test, and the tests that the simulations found oversized reject most often. The fixed-variance SPA test rejects in 5 windows, the Reality Check in 3, the two re-studentized tests in 4 and 4. The fixed-variance test rejects where the Reality Check does not in 2 windows, Dec 1960–Dec 1962 (0.047 against 0.072) and Jan 1971–Jan 1973 (0.049 against 0.063) (the two tests' p-values), and in one of them neither re-studentized test rejects. At block length 22, arch's unstudentized computation rejects in 6 windows against 3 at block length 8. LET rejects in only 2: its Šidák adjustment treats the twenty rules as independent, while rules with neighbouring lengths are almost the same rule, which makes it conservative, as in the correlated families of Section 3. Over the whole sample (26,117 days, mean block length 30, and 22 for the two tests marked so), the seven joint tests' p-values range from 0.031 to 0.050, at or just below the 5% level, while LET's is 0.316.

The illustration has the limits of its design. The rules are evaluated without transaction costs, so the series measure timing, not profit. The windows are not independent tests of one hypothesis: they are reported side by side, without a correction across windows, to show where the tests disagree, not to claim that any rule has skill. The one-sided test asks whether any rule beats buy-and-hold, and a rule that sits in the risk-free asset beats it in a falling market. Finally, the differences between tests come down to a handful of windows; the simulations, not this illustration, establish which test is calibrated.

Published predictors. Strategy research also searches over published cross-sectional predictors. A second illustration, pre-registered with three predictions before the data were analysed, applies Romano and Wolf's stepwise procedure to the monthly long-short returns of the predictors in the Open Source Asset Pricing data [13]. It uses 9 overlapping ten-year windows from 1975 to 2024, each containing every predictor with a full record (162 to 207), and asks how many predictors each test declares superior to zero (Table 12). The studentization decides much of the answer. Summed over the windows, the fixed-variance version declared 298 predictors superior at block length 5 and 348 at block length 10, arch's default, against 225 for the bootstrap-t: 32% and 55% more. The

Table 12: Published cross-sectional predictors declared superior to zero by Romano–Wolf stepwise tests at 5%, in ten-year windows of monthly long-short returns (pre-registered illustration; counts of predictors).

Window	Fixed, $q = 10$	Fixed, $q = 5$	Bootstrap-t	Reality Check
1975–1984 ($k = 162$)	45	43	27	8
1980–1989 ($k = 179$)	69	67	58	13
1985–1994 ($k = 188$)	61	53	42	22
1990–1999 ($k = 193$)	64	56	42	15
1995–2004 ($k = 202$)	46	40	32	1
2000–2009 ($k = 206$)	17	15	12	1
2005–2014 ($k = 206$)	17	9	5	1
2010–2019 ($k = 207$)	26	15	7	2
2015–2024 ($k = 195$)	3	0	0	3

Data: Open Source Asset Pricing release 2.0.0 [13], every predictor with a return in all 120 months of the window. Stationary bootstrap with 1,999 resamples shared by all tests; mean block length $q = 5$ except where stated. Fixed: Hansen's studentization, the Politis–Romano standard deviation held fixed in every resample (at $q = 10$, arch's default, this is the computation arch merged in October 2026). Bootstrap-t: each resample studentized by its own standard deviation. Totals over the windows: Fixed, $q = 10$ 348, Fixed, $q = 5$ 298, Bootstrap-t 225, Reality Check 66.

fixed-variance count was at least the bootstrap-t's in 9 of the 9 windows, and in 2005–2014 it was 17 at block length 10 against 5. All 3 predictions held. The unstudentized Reality Check declared far fewer (66), because without studentizing the maximum is dominated by the most volatile predictors, which hides precisely estimated ones. Like the timing rules, these are gross returns without trading costs; the counts measure how many predictors a test calls significant, not how many are profitable, and the overlapping windows are not independent.

7 Discussion

Implications for practice. Seven suggestions follow from the tables, within the limits below.

1. *Re-studentize inside the bootstrap, or do not studentize.* A studentized data-snooping test whose resampled statistics reuse the sample's variance estimate is oversized with block resampling at two years of daily data, more so with long blocks, many strategies or negative skew. Re-studentizing every resample, as Romano and Wolf [45] recommend, kept the size near the Reality Check's at block length 8 and keeps the power that studentizing buys when the skilled strategy is among the least volatile. arch adopts this for its next release [47]; users of arch 8.0 get an unstudentized test whatever they ask for, and users of the development code between 6 and 7 October 2026 got the fixed-variance one. With independent strategies and block length 8 the Reality Check and the bootstrap-t stayed within Monte Carlo error for 1 to 100 strategies, but every block-bootstrap test, the Reality Check included, was liberal with long blocks (block length 22) and up to 20 strategies, so the block length deserves as much care as the studentization. Under serial dependence the standard-deviation bootstrap-t was liberal, and re-studentizing by a natural block variance cured that case but was conservative elsewhere; re-studentizing every resample by its own Politis–Romano variance, as arch now does, treats the observed and resampled statistics alike and kept the size at 3.8–6.4% across the nine families at arch's default block length in an unregistered check (Table 8) and at 3.7–6.7% in the pre-registered confirmation (Table 10).

2. *Choose the ranking deliberately.* Studentizing, or ranking by t-statistic as LET does without resampling, finds a skilled low-volatility strategy that ranking by mean misses, and misses a skilled high-volatility strategy that ranking by mean finds.

3. *Recentre consistently and count ties.* Hansen's consistent recentring raises power at the nominal level when many variants are clearly poor; computing the p-value as $\#\{T^* \geq T\}/B$ stops a floored statistic from rejecting when no strategy beats the benchmark. The fix matters most for single-strategy calls and for searches with many poor variants.

4. *Check the selected strategy's returns for autocorrelation* before any multiple-testing adjustment, and do not wait for it to look large: at $\phi = 0.05$ the best-of-20 tests rejected 7.8% rather than 5%, and equation (1) gives the size for any ϕ, N and cutoff. Lo's factor removes most of the excess in the autoregressive case it assumes, though LET-Lo stayed about half a point liberal at small ϕ (Section 5); with correlated strategies, equation (3) gives the size instead, and the joint block-bootstrap tests, re-studentized or unstudentized, kept close to their level at $\phi \leq 0.1$. Kernel and block-bootstrap corrections reduce it without that assumption, but stay above the band (7.4% for HAC-t and 7.6% for SB-t) and are liberal even under i.i.d. returns (5.6% and 6.1%).

5. *Compare corrections at matched sides and with the number of resamples stated.* Otherwise the comparison measures conventions: the haircut's conservatism comes from its two-sided convention, and a bootstrap's power from its number of resamples.

6. *Treat* DSR ≥ 0.95 *as the demanding rule it is.* It asks whether the best strategy beats the expected best of N skill-less ones, not whether it has any skill, and over two years of daily data it passes few strategies with a true Sharpe ratio of 2.

7. *When returns are markedly skewed or fat-tailed, do not trust a single-series correction*: none in the benchmark is calibrated across negative skew, positive skew and fat tails, in part because selection itself distorts the moments of the chosen series. The Reality Check and the standard-deviation bootstrap-t, which resample all strategies together, stayed close to their level in those families.

Limitations. The nine families change one feature at a time; real returns combine several, and the only serial dependence studied is AR(1), the Lo factor's best case, at five levels for both the single-series and the joint tests. Proposition 2 assumes Gaussian returns, one autocorrelation shared by every strategy and a large-sample variance; part (b) adds one correlation shared by every pair of strategies, with factor and idiosyncratic parts equally persistent, and approximates the dependence of the t-statistics by that of their numerators. Real searches have unequal correlations, as in the block-cluster family, and unequal persistence; there the formula is an approximation, and by Slepian's inequality part (a) remains an upper bound only while every correlation is non-negative. Every null hypothesis is the least favourable configuration, in which all means are zero, except in the poor-alternative designs. The main grid covers one search shape ($N = 20$, $T = 504$) and one strongly skilled strategy; experiments B and E vary T, the number of strategies and the block length under one or two families only, and real searches often contain many weakly skilled strategies. Apart from experiments C and C2, which use i.i.d. normal returns, the strategies have equal unconditional variances. The joint tests use 1,000 resamples, and the block length of every resampling test is fixed by rule rather than chosen from the data [44]. Romano and Wolf's StepM was not run exactly as they specify it: the corrected stepwise test re-studentizes as they do but uses

a stationary bootstrap and Hansen's consistent recentring. The deflated Sharpe ratio is scored only at its 0.95 point, not as a probability. The external implementation of the studentized test, arch after its correction, and the author's corrected implementation were first scored in unregistered checks; the confirmation run v2c then tested them under a pre-registration, but it was written, run and scored by the same author with the same code, and its pre-registration, unlike the earlier ones, was not public before the run. It used 3,000 searches per cell for the joint tests, enough to separate a calibrated test from the fixed-variance one but not to distinguish sizes of 5% and 6%. Finally, a benchmark designed by the author of one of the scored methods, whose runs also chose defaults for the software of the author's firm, cannot be neutral in the sense of Bartoš et al. [6]. We pre-register the runs and publish every generator, validator, seed and result file, but independent replication with independently chosen families would be more convincing than anything we can do ourselves.

Next versions. The benchmark is versioned so that it can grow without rewriting its history. Natural additions are families that combine features (skewed and autocorrelated returns) or have non-AR dependence, unequal correlations and persistence in Proposition 2, alternatives with many weakly skilled strategies, data-driven block lengths, Romano and Wolf's StepM as specified, and an estimate of the effective number of trials for correlated searches.

8 Conclusion

The corrections practitioners use for backtest overfitting and data snooping keep their promises in the setting they were derived for and fail in predictable ways outside it. The most consequential failure found here is the most technical: with block resampling, Hansen's studentized SPA test, recommended over the Reality Check for its power, is oversized at the two-year horizon of strategy research, because its bootstrap reuses a noisy, downward-biased variance estimate. The remedy is simple and already in use: re-studentizing every resample, as Romano and Wolf recommend, kept the size near the Reality Check's at block length 8, and arch adopted it after the fixed-variance version was shown to be oversized. The remedy does not remove the second lesson of the joint tests: long blocks make every block-bootstrap test liberal, because the bootstrap's own variance is the same noisy estimate. Autocorrelation remains the most damaging violation for the single-series corrections, and a best-of-N cutoff makes even a lag-one autocorrelation of 0.05 matter, with independent or correlated strategies; a pre-registered confirmation on fresh seeds reproduced both this and the effect of re-studentizing. Several apparent differences between methods come from conventions—sides, thresholds, numbers of resamples, ties—rather than from the methods. An open, pre-registered benchmark with known ground truth makes such statements checkable, and anyone can rerun or extend it from the published code and seeds.

Code and data availability

All code, result files, pre-registrations and the source of this paper are at `https://github.com/arhancanli/null-zoo` (MIT licence); version 1 of the paper is archived at `https://doi.org/10.5281/zenodo.23018988`. The JavaScript benchmark in `src/` is copied unchanged from `https://github.com/arhancanli/canlicapital` at commits `4f7838f8db35` (v0), `ea3985981ec8` (v1, v1b) and `46ee1f2bca34` (the autocorrelation levels, whose result file we re-ran and reproduced byte for byte), where it is maintained. The corrected data-snooping implementation is copied unchanged from commit `4c7e429e7602` into `analysis/snooping-fix/`, with the script that compares it with the v1 code; the checks of arch's code are `analysis/v2/arch_merged_check.py`

and `analysis/v2/arch_882_check.py`. The confirmation run v2c—pre-registration, closed-form predictions, seed checks, run and scoring scripts, logs and results—is in `analysis/confirm/`. The second implementation, its instrument tests, the v2 and v2b pre-registrations, the run and scoring scripts, the empirical illustration and their outputs are in `analysis/`. Every result in the text and tables is generated from the result files by `analysis/tables.mjs`, and the paper reads each one by key; design constants and figures quoted from the literature are typed.

Pre-registration record. Each pre-registration was committed before its run, and those of v1–v2b were also pushed (commit, time): v1 `b3102e90`, 2026-10-04T05:09:00Z, results `412f1907`, 2026-10-04T05:39:46Z; v1b `abbf2dae`, 2026-10-04T05:41:10Z, results `5c0926b8`, 2026-10-04T06:28:31Z (both in https://github.com/arhancanli/canlicapital/pull/379); v2 `7658b3b46ddb`, 2026-10-05T15:43:53Z, run started 2026-10-05T15:44:17Z; v2b `ea2d963e3825`, 2026-10-05T18:01:36Z, data downloaded 2026-10-05T18:01:55Z, run started 2026-10-05T18:04:22Z; v2c `c0996a70714f`, 2026-11-07T18:24:34Z, run started 2026-11-07T18:26:58Z, a signed commit that, unlike the others, was not pushed before the run. The times are those recorded by git and GitHub; no third-party timestamp was obtained.

Reproduction. The published v0 result file (SHA-256 `e8287f19cf25a7e0858ab12d19e2c2154f1068c5c32459123d68ec3a7cd3ce0f`) reproduces byte for byte. Continuous integration checks that reproduction, the instrument tests of both implementations, and that the committed evaluations and tables match the result files; it does not re-run v1, v1b, v2 or v2b, which take hours. We re-ran v1 with 4 worker threads instead of 12, and v2 with 6 worker processes instead of 8; both result files were byte-identical (SHA-256 `433166b4af950cd586c4b3941de8b4c4c18bde55833087e4bd30e8346229f031` and `15a0fa993d142a0863f3ef39c56b29c3b1626addc24b46caadf31c44fc72774d`). The Python runs used numpy 2.5.3; numpy does not promise identical random streams across versions, so a byte-for-byte re-run needs that version. The empirical illustration downloads its data from the French Data Library, whose files are revised over time; the script refuses a file whose SHA-256 differs from the one recorded (`2f29e22546069914890a712680a6f81f3680ebc3543b52864484d209ca13a7db`, downloaded 2026-10-05T18:01:55Z).

Competing interests

The author designed the luck-equivalent-trials statistic scored here. The author's firm, Canli Capital, publishes a validation API, free at the time of writing, and MCP servers that implement the luck-equivalent-trials test, the haircut, the deflated Sharpe ratio and the data-snooping tests; the v1 and v1b runs set their defaults (the Reality Check for searches, LET-Lo for single series), and the firm may benefit from wider use of these tools. No external funding was received.

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A Proofs

Proof of Lemma 1. The estimate $\hat{\omega}^2$ is unchanged when a constant is added to every x_t, so take $E[x_t] = 0$.

(a) With δ_{ts} the Kronecker delta, $E[(x_t - \bar{x})(x_s - \bar{x})] = \sigma^2(\delta_{ts} - 1/T)$. Hence $E[\hat{\gamma}_0] = \sigma^2(T-1)/T$ and, for $i \geq 1$, $E[\hat{\gamma}_i] = -\sigma^2(T-i)/T^2$. Summing with weights 1 and $2\kappa_i$ gives $E[\hat{\omega}^2] = \sigma^2 r_T(q)$.

(b) Under normality the sample mean is independent of the vector of deviations $(x_t - \bar{x})_t$, and $\hat{\omega}^2$ is a function of the deviations alone. For the variance, let $\tilde{\gamma}_i = T^{-1}\sum_{t=1}^{T-i} x_t x_{t+i}$. For independent mean-zero normal variables, $E[x_t x_{t+i} x_s x_{s+j}]$ with $i, j \geq 0$ is non-zero only when the four indices pair up. This gives $\mathrm{Var}(\tilde{\gamma}_0) = 2\sigma^4/T$, $\mathrm{Var}(\tilde{\gamma}_i) = \sigma^4(T-i)/T^2$ for $i \geq 1$, and $\mathrm{Cov}(\tilde{\gamma}_i, \tilde{\gamma}_j) = 0$ for $i \neq j$, so $\mathrm{Var}(\tilde{\gamma}_0 + 2\sum_i \kappa_i \tilde{\gamma}_i) = \sigma^4 v_T(q)$. The difference $\hat{\gamma}_i - \tilde{\gamma}_i$ consists of terms in $\bar{x}$ times partial

sums of the x_t. For fixed q, where $\sum_i \kappa_i$ is bounded, these change $\hat{\omega}^2$ by $O_p(T^{-1})$, and they change its variance by $O(T^{-2})$.

(c) For each i, $\kappa_i = (1-i/T)(1-p)^i + (i/T)(1-p)^{T-i}$ is non-negative and decreasing in $p = 1/q$, so it increases with q. Both $1 - r_T(q)$ and $v_T(q)$ are increasing functions of the κ_i. □

Proof of Proposition 1. By Lemma 1(b), $Z_k = \sqrt{T}\,\bar{x}_k/\sigma_k$ is standard normal and independent of $\hat{\omega}_k$, and the N pairs $(Z_k, \hat{\omega}_k)$ are independent. With $\hat{\omega}_k^2/\sigma_k^2 = r_T W_k$ and $W_k \sim \chi^2_\nu/\nu$, each statistic is $Z_k/\sqrt{r_T W_k} = t_k/\sqrt{r_T}$, with t_k distributed as Student's t with ν degrees of freedom. Then $\Pr(\max_k t_k/\sqrt{r_T} > z^*) = 1 - [1 - \bar{F}_\nu(z^*\sqrt{r_T})]^N$, which is (5).

With ϕ and $\bar{\Phi}$ the standard normal density and upper tail, for $c > 0$ let $g_c(w) = \bar{\Phi}(c\sqrt{w})$, so that $\bar{F}_\nu(c) = E[g_c(W)]$ with $W \sim \chi^2_\nu/\nu$. Differentiating twice gives

$$g_c''(w) = \frac{c\,\phi(c\sqrt{w})}{4w^{3/2}}\,(1 + c^2 w) > 0,$$

so g_c is convex on $w > 0$. Since $E[W] = 1$, Jensen's inequality gives $\bar{F}_\nu(c) \geq \bar{\Phi}(c)$.

For $\nu_1 < \nu_2$, the gamma–beta decomposition gives $W_{\nu_1} \overset{d}{=} W_{\nu_2} M$, with $M = (\nu_2/\nu_1)B$, $B \sim \text{Beta}(\nu_1/2, (\nu_2 - \nu_1)/2)$ independent of W_{ν_2}, and $E[M] = 1$. Conditioning on W_{ν_2} and applying Jensen's inequality to the convex function $m \mapsto g_c(W_{\nu_2} m)$ gives $\bar{F}_{\nu_1}(c) \geq \bar{F}_{\nu_2}(c)$. So the upper tail at any $c > 0$ decreases with the degrees of freedom. It also decreases with c.

By Lemma 1(c), a longer block lowers r_T, which lowers the argument $z^*\sqrt{r_T}$, and raises v_T, which lowers ν. Both raise $\bar{F}_\nu(z^*\sqrt{r_T})$ and therefore $\pi_T(q)$. Because $r_T < 1$ and $\nu < \infty$, $\bar{F}_\nu(z^*\sqrt{r_T}) > \bar{\Phi}(z^*) = 1 - (1-\alpha)^{1/N}$, so $\pi_T(q) > \alpha$. As $T \to \infty$ with q fixed, $r_T \to 1$ and $\nu \to \infty$, so $\bar{F}_\nu(z^*\sqrt{r_T}) \to \bar{\Phi}(z^*)$ and $\pi_T(q) \to \alpha$. □

Proof of Proposition 2(a). Write $\lambda = (1+\phi)/(1-\phi)$. For a stationary AR(1) with $|\phi| < 1$, innovation variance $\sigma^2(1-\phi^2)$ and hence marginal variance σ^2, the autocovariances are $\sigma^2\phi^{|h|}$, so

$$\text{Var}(\sqrt{T}\,\bar{x}) = \frac{\sigma^2}{T}\sum_{|h|<T}(T - |h|)\,\phi^{|h|} = \sigma^2\Big[\lambda - \frac{2\phi(1-\phi^T)}{T(1-\phi)^2}\Big] = \sigma^2\lambda + O(T^{-1}).$$

Under Gaussianity $\sqrt{T}\,\bar{x}/\sigma$ is exactly normal with this variance, and $s^2 \to \sigma^2$ in probability (ergodicity), so $t = \sqrt{T}\,\bar{x}/s$ converges in distribution to $N(0, \lambda)$ by Slutsky's lemma. Replacing that limit by $\sqrt{\lambda}$ times a Student t_{T-1} variable—which is exact at $\phi = 0$ and differs from the normal limit by $O(T^{-1})$ in the tail at a fixed cutoff—gives $\Pr(t_k > c) \approx \bar{F}_{T-1}(c/\sqrt{\lambda})$ for each strategy. With independent strategies, the probability that none of the N exceeds c is the product of the N marginal probabilities, which gives (1). Because $\bar{F}_{T-1}$ is decreasing and $\lambda > 1$ for $\phi > 0$, the size exceeds the i.i.d. size and increases with ϕ.

For the elasticity, let $s = \sqrt{\lambda}$ and $u(s) = \bar{F}_{T-1}(c/s)$, so $\pi = 1 - (1-u)^N$. Then $\mathrm{d}\ln u/\mathrm{d}\ln s = c f_{T-1}(c)/\bar{F}_{T-1}(c)$ at $s = 1$, with f_{T-1} the density, and $\mathrm{d}\ln\pi/\mathrm{d}\ln u = Nu(1-u)^{N-1}/\pi$, which is close to one when π is small. The product is (2). The ratio $cf(c)/\bar{F}(c)$ is c times the hazard rate of the distribution, which for the normal distribution exceeds c^2 and grows without bound with c; a best-of-N test uses a larger cutoff than a single test at the same level, so the same proportional error in the standard error moves its size by a larger proportion. □

Proof of Proposition 2(b). Let $\lambda_T = \text{Var}(\sqrt{T}\,\bar{x}_k)/\sigma^2$, the bracket above. Under the factor construction, $\text{Cov}(x_{j,t}, x_{k,s}) = \rho\,\sigma^2\phi^{|t-s|}$ for $j \neq k$, so the same double sum gives

$$\text{Cov}(\sqrt{T}\,\bar{x}_j, \sqrt{T}\,\bar{x}_k) = \frac{\rho\,\sigma^2}{T}\sum_{|h|<T}(T - |h|)\,\phi^{|h|} = \rho\,\sigma^2\lambda_T.$$

Hence the vector $Y = (\sqrt{T}\,\bar{x}_k/(\sigma\sqrt{\lambda_T}))_{k=1}^N$ is exactly Gaussian with unit variances and every pairwise correlation equal to ρ, for every T; the autocorrelation scales the variance but leaves the correlation unchanged. For $\rho \geq 0$ such a vector can be written $Y_k = \sqrt{\rho}\,Z_0 + \sqrt{1-\rho}\,Z_k$ with $Z_0, \ldots, Z_N$ independent standard normal [54, Ch. 8]. Conditional on $Z_0 = z$ the Y_k are independent, so

$$\Pr\Big(\max_k Y_k \leq y\Big) = \int_{-\infty}^{\infty} \Phi\Big(\frac{y - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\Big)^N \varphi(z)\,\mathrm{d}z,$$

Each $s_k \to \sigma$ in probability, so by Slutsky's lemma the vector of t-statistics converges in distribution to $\sqrt{\lambda}\,Y$ (with $\lambda_T \to \lambda$), with φ the standard normal density, and the test's size converges to one minus the display at $y = c/\sqrt{\lambda}$. Part (a) keeps the Student t_{T-1} marginal; to keep it here, the cutoff is moved to the normal scale through the marginal tail, $y = c' = \bar{\Phi}^{-1}(u)$ with $u = \bar{F}_{T-1}(c/\sqrt{\lambda})$, so that each $\Pr(Y_k > c') = u$ exactly. This gives (3). It is a Gaussian-copula approximation: it treats the dependence of the t-statistics as that of their numerators, ignoring the (asymptotically negligible) dependence between the s_k. At $\rho = 0$ the integrand is $\Phi(c')^N = (1-u)^N$, which is (1). Slepian's inequality [51] states that for a centred Gaussian vector with fixed variances $\Pr(\max_k Y_k \leq y)$ does not decrease when any correlation increases; so (3) is non-increasing in ρ and, for $\rho \geq 0$, at most (1). As $\rho \to 1$ the Y_k coincide and the size tends to u, a single test at the best-of-N cutoff. □

B Additional results

B.1 The deflated Sharpe ratio's hurdle

Formally, the rule tests $H_0\colon SR \leq SR_0$, where SR_0 is the expected maximum of N skill-less trials, a harder hypothesis than "the best strategy has no skill". If $1 - \mathrm{DSR}$ were a calibrated p-value for no skill, a value of at least 0.95 would occur in about 5% of skill-less searches; it occurred in at most 1.7%, so at its 0.95 point the rule is strongly conservative for that hypothesis. The difference in hurdles is large:

- Under independence and normality, the one-sided 5% critical value for the best of $N = 20$ trials is about 2.80 standard errors.
- The DSR's benchmark is about 1.90 cross-trial standard deviations (the approximation of Bailey et al. [4]; the exact expected maximum of 20 standard normals is 1.87), and the rule asks for a further 1.645 standard errors on top of that.
- The cross-trial standard deviation is computed from all 20 strategies, including the best. In the (unregistered) v0-design robustness run it averaged 0.99 times the i.i.d. standard error under the null, a hurdle of about 3.53 standard errors; under the alternative the skilled strategy widens the dispersion to 1.17 times the standard error and the hurdle to about 3.87: a skilled strategy raises its own benchmark.
- When strategies are strongly correlated the dispersion shrinks; below 0.61 times the standard error the DSR's hurdle falls under the Šidák value. In the correlated family it was 0.70, a hurdle of about 2.97, which is why the DSR rejects most there.

The ranking the DSR induces is better than its raw power suggests: size-adjusted, its power under i.i.d. returns is 40.4% against 53.2% for the Šidák test, and across families it trails by 5.6–17.0 points. Under i.i.d. returns the DSR's raw power falls 44.0 points short of the Šidák test's size-adjusted power; 31.2 of those points come from the 0.95 threshold (the DSR's own size-adjusted

power is that much higher than its raw power), and the remaining 12.8 from the statistic, whose benchmark moves with the dispersion of the whole search.

B.2 Resolution matters for the bootstrap

A bootstrap p-value with B resamples is a Monte Carlo estimate, and a test built on it loses power as B falls [16]. With a Šidák adjustment the loss is sharp: with 400 resamples the smallest attainable adjusted p-value is 0.0487, just below the level, so the test rejects only when no resample reaches the observed Sharpe ratio. Scoring the same 10,000 searches per cell with 400 and with 1,999 resamples in an unregistered paired run (Table 22), the bootstrap's power under i.i.d. normal returns rises from 46.7% to 50.9% (the Šidák t test: 52.6%), while its size moves from 4.28% to 4.50%. Comparisons of resampling tests with analytical ones should state the number of resamples.

B.3 The Harvey–Liu haircut read one-sided is the Šidák test

For a positive Sharpe ratio, the two-sided $p \leq 0.05$ of Harvey and Liu [28] is a one-sided test at 2.5%, which is why the haircut with its authors' convention sits near half the level (2.2% under i.i.d. normal returns). Read one-sided, the haircut and the luck-equivalent-trials test use the same Student t tail p_1 of the same statistic; Bonferroni rejects when $Np_1 \leq 0.05$ and Šidák when $1-(1-p_1)^N \leq 0.05$, and the two disagree only when p_1 lies between 0.002500 and 0.002561. Their sizes differ by at most 0.36 points in any family. This is a known equivalence, not a finding; its practical consequence is that comparisons of these corrections at unmatched sides measure the choice of sides.

B.4 The bootstrap's own variance

The bootstrap's critical value carries the same estimate. Under the stationary bootstrap the conditional variance of a resampled mean is the Politis–Romano estimate, so every block-bootstrap test, studentized or not, inherits that noisy, downward-biased estimate through its critical value, and the maximum over many strategies averages part of the noise out. Experiment E of v2b, registered after the v2 results, measures how much this matters under i.i.d. normal returns (Table 13). With one strategy, the Reality Check rejected 5.2% of skill-less searches at block length 1, 5.5% at 8 and 6.5% at 22 (P11 held). At block length 8 the effect is small: the Reality Check and the bootstrap-t stayed within the Monte Carlo band for every number of strategies from 1 to 100 (5.0–5.6% and 5.1–5.6%), and P12, which expected the Reality Check's size with one strategy to exceed its size with a hundred by at least half a point at block length 8, failed: the difference was 0.48 points. At block length 22 both were liberal with up to 20 strategies (6.4–6.6% and 6.3–6.7%), and the maximum over 50 or 100 strategies damped the effect (5.3% and 4.7% for the Reality Check). The fixed-variance test compounds the two effects. At block length 8 its size rose from 5.5% with one strategy to 9.1% with a hundred, and at block length 22 from 6.5% to 20.5%; at block length 1, where its variance is the sample variance, it stayed at 5.1–5.7%. With twenty strategies and block length 8 it rejected 7.5% of searches, a fourth replication of the main result on fresh seeds (P10 held).

Two v2 observations that pointed to the number of strategies were not replicated. With five i.i.d. strategies at block length 8, experiment B had found 6.0% for the Reality Check and 6.2% for the bootstrap-t; experiment E, on fresh seeds, found 5.0% and 5.2%. In the correlated and cluster families, however, the Reality Check and the bootstrap-t rejected 5.4–6.2% of skill-less searches across the runs, at or just above the upper edge of the band, which the number of independent strategies alone does not explain. P2's failure in v2, the sample-standard-deviation version at 6.4% in the cluster family, belongs to that pattern.

B.5 Re-studentizing with a block variance

Re-studentizing every resample by its own standard deviation ignores serial dependence, and under AR(1) returns that bootstrap-t was liberal (6.6% in v2, 6.9% in v2b). Experiment F re-studentized each resample instead by its natural block variance—the squared sums, within each resampled block, of the deviations from the resample mean, averaged over the T periods [24]—while the observed statistic kept the Politis–Romano studentization (Table 14). Under AR(1) returns this brought the size to 5.5% (P14 held), and outside the AR(1) family it never exceeded 6.5% (P13 held). But it over-corrects: outside the AR(1) family its size was 3.3–4.4%, below the band in all eight families, and experiment E shows it growing more conservative with more strategies and longer blocks (3.2% with a hundred strategies at block length 8, 1.0% at 22). Its size-adjusted power was close to the standard-deviation version's (50.3% against 51.8% under i.i.d. normal returns), so what it loses is level, not ranking. A likely cause, which we have not tested, is that the observed and the resampled statistics are now studentized by different estimators: the natural block variance of a resample, computed from blocks of random length, is noisier than the sample's Politis–Romano estimate, which spreads the bootstrap distribution of the maximum and raises its critical value. A block studentization that treats the observed and the resampled statistics alike was not part of any registered run; Gonçalves and Vogelsang [23] is a natural starting point, and arch's corrected test, which re-studentizes every resample by its own Politis–Romano variance, is one such studentization (Section 4.5, unregistered).

B.6 Where studentizing and recentring pay off

Studentizing exists for a reason, and experiment C2 shows when it helps and when it hurts (Table 15). With volatilities from 0.5 to 2, the strategy with the largest mean was the skilled one in 17.1% of searches when the skilled strategy was the least volatile, 66.3% when it had the median volatility and 93.7% when it was the most volatile; the strategy with the largest t-statistic was the skilled one in about four searches in five wherever it sat (81.1%, 80.5% and 80.2%). The Reality Check, which ranks by mean, accordingly rejected in 5.4%, 18.1% and 78.4% of searches with a Sharpe ratio of 2, and the bootstrap-t in 53.6%, 54.0% and 53.4%: studentizing pays when the skilled strategy is among the less volatile and costs power when it is the most volatile (P15 held). The gain comes from ranking by t-statistic, not from the bootstrap: LET, the Šidák t test, which does not resample, had about the bootstrap-t's power (52.9%, 53.2% and 52.5%). The fixed-variance version was again oversized (7.1%), and the natural-block version again conservative (3.6%).

When the other strategies are clearly poor, the Reality Check, which recentres every strategy at its mean, is very conservative (size 0.2%, power 50.9%), and Hansen's consistent recentring raises power at the nominal level to 87.2%, with or without studentizing, once ties are counted (sizes 5.0%, 5.0% and 4.9%). Size-adjusted, the Reality Check ranks these searches as well as any test (88.7%): its loss is conservativeness at the nominal level, which is the point of Hansen [27]. In the v2 version of this design (Table 20), with the strict inequality, prediction P8 compared the raw power of an oversized bootstrap-t (size 7.3%) with that of the conservative Reality Check; it held at the nominal level, not after size adjustment. Studentizing and recentring do what Hansen proposed them for. The lessons are to studentize inside the bootstrap, to count ties, and to remember that studentizing changes which strategies a test can find.

Table 13: Size against the number of strategies k and the mean block length (1, 8 or 22), Null Zoo v2b, experiment E: i.i.d. normal returns, $T = 504$, 5,000 null searches per cell.

	RC			SPA_c			SPA_c-BT			SPA_c-NB		
	1	8	22	1	8	22	1	8	22	1	8	22
$k = 1$	5.2	5.5	**6.5**	5.2	5.5	**6.5**	5.1	5.4	**6.5**	5.1	4.8	4.8
$k = 2$	5.5	5.5	**6.4**	5.7	5.8	**7.4**	5.7	5.6	**6.7**	5.7	4.8	4.4
$k = 5$	4.9	5.0	**6.6**	5.1	**5.9**	**9.2**	4.9	5.2	**6.5**	4.9	*4.2*	*4.1*
$k = 10$	5.3	5.6	**6.4**	5.5	**7.0**	**11.2**	5.3	5.6	**6.5**	5.3	4.5	*3.0*
$k = 20$	5.0	5.2	**6.4**	5.3	**7.5**	**12.7**	5.0	5.1	**6.3**	5.0	*4.0*	*2.5*
$k = 50$	4.9	5.0	5.3	5.6	**8.1**	**16.9**	5.3	5.1	5.6	5.3	*3.6*	*1.5*
$k = 100$	4.8	5.0	4.7	5.6	**9.1**	**20.5**	5.2	5.1	5.0	5.2	*3.2*	*1.0*

Bold: above the 99% Monte Carlo band around 5% (4.21–5.79 for 5,000 searches); italic: below it. Floored tests count ties. At block length 1 the Politis–Romano variance is the sample variance, so SPA_c is then SPA_c-SD. SPA_c-NB re-studentizes every resample by its natural block variance. The Reality Check and SPA_c-U differ by at most 0.2 points in any cell; SPA_c-U and the strict-inequality SPA_c are in Table 23.

Table 14: Re-studentizing every resample, by its own standard deviation (SPA_c-BT) or by its natural block variance (SPA_c-NB), in the nine families: Null Zoo v2b, experiment F (mean block length 8; 10,000 null and 5,000 skill searches per family; size and size-adjusted power at a Sharpe ratio of 2, percent).

	Size				Size-adjusted power		
Family	RC	SPA_c	SPA_c-BT	SPA_c-NB	RC	SPA_c-BT	SPA_c-NB
IID normal	5.3	**7.5**	5.5	*4.0*	52.4	51.8	50.3
Student t_4	4.6	**7.5**	5.3	*4.1*	52.0	52.9	50.2
Skew -1.32	5.2	**12.7**	5.5	*4.1*	55.0	46.0	44.1
Skew $+1.32$	5.1	*4.1*	5.5	*3.8*	51.0	59.5	58.7
GARCH(1,1)	4.8	**6.2**	4.8	*3.3*	51.1	54.1	54.2
AR(1), $\phi = 0.2$	**6.6**	**9.7**	**6.9**	5.5	34.3	33.8	31.6
Two vol. regimes	4.8	**6.0**	5.4	*3.4*	50.6	53.5	54.5
Correlated, $\rho = 0.5$	**5.7**	**7.0**	**5.7**	*4.3*	55.6	55.0	53.5
Clusters, $\rho = 0.8$	**5.6**	**7.1**	**5.6**	*4.4*	58.5	57.6	55.7

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. Floored tests count ties. Raw power and the other variants are in Table 24.

C Additional tables

Table 16 gives the size of the validators not shown in the main tables and of the other joint tests in the confirmation run v1b. Tables 17 and 18 give raw power at Sharpe ratios of 1 and 3. Table 19 gives the power of the SPA variants of experiment A, Table 20 the v2 version of experiment C (strict inequality), Table 21 the check of arch itself (experiment D), and Table 22 the unregistered bootstrap-resolution run. Tables 23 and 24 complete experiments E and F of v2b, Table 25 lists every window of the empirical illustration, and Tables 26, 27 and 28 list the pre-registered predictions of v2 and v2b and the hypotheses of v2c with their outcomes.

Table 15: Where studentizing and recentring matter, with ties counted (Null Zoo v2b, experiment C2; i.i.d. normal returns, 504 periods, 20 strategies, 10,000 searches per cell; percent).

	Unequal volatilities				Poor alternatives	
Test	Null	Skill low	Skill median	Skill high	Null	Skill
RC	5.4	5.4 / 5.0	18.1 / 16.8	78.4 / 77.5	*0.2*	50.9 / 88.7
SPA_c-U	5.6	5.6 / 5.2	18.8 / 17.3	78.5 / 77.7	5.0	87.2 / 87.2
SPA_c (>)	**7.1**	56.6 / 50.8	57.2 / 51.5	56.8 / 51.2	**7.4**	87.2 / 82.3
SPA_c	**7.1**	56.6 / 50.8	57.2 / 51.5	56.8 / 51.2	5.0	87.2 / 87.2
SPA_c-BT	4.9	53.6 / 53.8	54.0 / 54.3	53.4 / 53.7	4.9	87.2 / 87.3
SPA_c-NB	*3.6*	46.6 / 50.3	47.3 / 51.1	46.9 / 50.6	*4.4*	85.7 / 87.1
LET	4.5	52.9 / 54.1	53.2 / 54.5	52.5 / 54.0	*0.2*	50.1 / 88.7

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. Null columns: size; skill columns: raw / size-adjusted power. Unequal volatilities: $0.5 \cdot 4^{j/19}$ for strategy $j = 0, \ldots, 19$ (0.5 to 2), all Sharpe ratios 0 under the null; under the alternative strategy 0 (low, the least volatile), 10 (median) or 19 (high, the most volatile) has a Sharpe ratio of 2. Poor alternatives: unit volatility, 19 strategies with a Sharpe ratio of −3, strategy 0 at 0 (null) or 2. SPA_c (>) uses Hansen's strict inequality; the other floored tests count ties. LET is the Šidák-adjusted t test on the best Sharpe ratio.

Table 16: Size of the remaining v1 validators, and of the other joint tests in the confirmation run v1b (percent, 10,000 searches per cell).

Family	HL 2s	HL-Lo	Holm	Oracle	SPA_u v1b	Step-SPA v1b
IID normal	2.2	5.0	4.8	**5.6**	**7.5**	**7.7**
Student t_4	2.3	5.1	4.6	5.4	**7.5**	**7.6**
Skew −1.32	4.4	**8.5**	**8.1**	**5.9**	**12.3**	**12.6**
Skew +1.32	1.3	*3.3*	*2.9*	4.5	*4.0*	*4.3*
GARCH(1,1)	2.3	4.9	4.9	5.1	**6.2**	**6.3**
AR(1), $\phi = 0.2$	11.9	5.3	**19.3**	**5.7**	**8.9**	**9.1**
Two vol. regimes	2.7	5.1	5.0	5.3	**6.0**	**6.1**
Correlated, $\rho = 0.5$	1.9	*3.7*	*3.5*	5.4	**6.9**	**6.9**
Clusters, $\rho = 0.8$	1.6	*3.4*	*3.3*	5.1	**6.9**	**7.0**

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it. HL 2s: the haircut with the two-sided p-values its authors define; its nominal one-sided level is 2.5%, so it is not flagged.

Table 17: Raw power at a true annualized Sharpe ratio of 1 (percent), Null Zoo v1.

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC-t	RC	SPA_c
IID normal	41.2	12.4	13.0	0.4	12.1	13.7	13.1	16.8
Student t_4	41.9	13.2	13.8	0.6	14.8	14.3	12.4	17.0
Skew −1.32	42.5	16.8	17.3	0.5	11.7	18.4	14.0	23.6
Skew +1.32	41.1	9.7	10.2	0.4	13.3	10.8	12.5	11.6
GARCH(1,1)	42.7	13.6	13.7	0.5	12.7	14.3	12.4	16.7
AR(1), $\phi = 0.2$	33.4	29.9	10.1	0.9	28.4	13.4	11.6	15.6
Two vol. regimes	42.7	13.8	14.2	0.6	13.6	14.2	11.4	15.8
Correlated, $\rho = 0.5$	41.5	10.1	10.4	4.0	9.6	11.1	14.6	16.4
Clusters, $\rho = 0.8$	41.3	11.0	11.4	0.7	10.7	12.1	16.0	18.1

Table 18: Raw power at a true annualized Sharpe ratio of 3 (percent), Null Zoo v1.

Family	Oracle	LET	LET-Lo	DSR	Boot	HAC-t	RC	SPA$_c$
IID normal	99.4	93.0	92.8	50.6	92.2	93.0	93.1	93.6
Student t_4	99.5	92.1	92.0	49.9	90.6	92.2	91.9	92.9
Skew -1.32	99.3	91.3	91.2	41.1	86.8	91.4	93.6	93.1
Skew $+1.32$	99.8	94.3	94.1	60.1	95.4	94.3	92.0	94.0
GARCH(1,1)	99.4	92.4	92.3	52.3	91.5	92.5	92.3	93.1
AR(1), $\phi = 0.2$	97.2	90.1	75.9	37.7	89.3	79.4	78.4	80.6
Two vol. regimes	99.3	91.6	91.5	53.1	90.6	91.4	91.2	91.6
Correlated, $\rho = 0.5$	99.5	92.4	92.1	65.6	91.6	92.5	94.7	95.1
Clusters, $\rho = 0.8$	99.5	93.1	92.9	59.9	92.1	93.1	95.5	95.5

Table 19: Power (raw / size-adjusted) of the SPA variants at a true annualized Sharpe ratio of 2, Null Zoo v2 experiment A.

Family	SPA$_c$	SPA$_c$-SD	SPA$_c$-BT	RC	SPA$_c$-U
IID normal	57.2 / 50.9	54.6 / 53.0	53.9 / 53.4	53.8 / 53.2	54.2 / 53.2
Student t_4	57.5 / 52.6	54.8 / 54.6	54.8 / 54.5	51.6 / 52.7	52.0 / 53.1
Skew -1.32	60.9 / 44.1	58.6 / 46.8	48.1 / 46.7	55.7 / 54.9	56.1 / 54.8
Skew $+1.32$	53.0 / 55.7	49.6 / 58.2	59.2 / 57.8	49.7 / 49.1	50.0 / 49.4
GARCH(1,1)	58.4 / 54.6	55.3 / 54.9	55.0 / 54.3	50.7 / 50.7	51.1 / 50.8
AR(1), $\phi = 0.2$	42.8 / 31.6	39.8 / 34.2	39.0 / 34.2	38.7 / 34.8	39.2 / 34.8
Two vol. regimes	58.1 / 56.1	54.7 / 54.9	54.8 / 54.8	49.3 / 50.4	49.6 / 50.5
Correlated, $\rho = 0.5$	59.9 / 54.8	58.8 / 56.5	58.1 / 56.5	58.2 / 56.5	58.4 / 56.7
Clusters, $\rho = 0.8$	61.9 / 53.8	60.8 / 55.6	60.2 / 55.6	60.1 / 56.3	60.3 / 56.1

Table 20: Unequal volatilities with the skill on the least volatile strategy, and poor alternatives, with the strict inequality (Null Zoo v2, experiment C; i.i.d. normal returns, 504 periods, 20 strategies, 10,000 searches per cell). Size / power / size-adjusted power, in percent.

Test	Unequal volatilities	Poor alternatives
RC	5.5 / 5.0 / 4.4	*0.2* / 51.4 / 88.9
SPA$_c$-U	**5.7** / 5.2 / 4.3	4.8 / 87.3 / 87.7
SPA$_c$	**7.6** / 56.7 / 48.9	**7.4** / 87.4 / 82.7
SPA$_u$	**7.5** / 56.4 / 49.0	*0.3* / 53.3 / 88.3
SPA$_c$-SD	5.6 / 53.8 / 52.2	**7.4** / 87.3 / 82.4
SPA$_c$-BT	5.2 / 53.1 / 52.4	**7.3** / 87.1 / 82.2
LET	4.8 / 52.4 / 53.0	*0.2* / 50.7 / 88.8

The designs of Table 15, before ties were counted: every floored test here uses the strict inequality $p = \#\{T^* > T\}/B$.

Table 21: arch 8.0.0 on Null Zoo searches (experiment D, 2,000 null searches per family): size of arch's consistent and upper p-values and of this paper's implementation on the same searches, at mean block length 8 and at arch's default (22).

Family	arch_c (8)	SPA_c-U (8)	arch_u (8)	RC (8)	arch_c (22)	SPA_c-U (22)	SPA_c (22)
AR(1), $\phi = 0.2$	**7.8**	**7.6**	**7.7**	**7.4**	**7.4**	**7.7**	**13.9**
IID normal	5.7	6.0	5.6	5.8	**6.5**	**6.5**	**12.7**
Skew −1.32	5.1	5.0	5.0	4.7	5.5	5.5	**17.9**

Bold: above the 99% Monte Carlo band around 5% (3.74–6.26 for 2,000 searches); italic: below it. Searches on which arch's p-values differed between `studentize=True` and `studentize=False`: 0 of 12,000 search-and-block pairs.

Table 22: Bootstrap resolution: the same 10,000 searches per cell scored by the single-series bootstrap with 400 and with 1,999 resamples (v0 design, robustness generator; not pre-registered). Size and power in percent.

Family	LET size	Boot_{400} size	Boot_{1999} size	LET power	Boot_{400} power	Boot_{1999} power
IID normal	4.59	4.28	4.50	52.6	46.7	50.9
Student t_4	4.80	5.86	6.24	54.1	49.0	53.9
Skew −1.32	8.02	4.89	5.26	54.9	40.9	46.0

Table 23: The two variants of experiment E not shown in Table 13: arch's computation (SPA_c-U) and Hansen's fixed-variance test with the strict inequality (SPA_c (>)); size in percent.

	SPA_c-U			SPA_c (>)		
	1	8	22	1	8	22
$k = 1$	5.2	5.5	**6.5**	**8.0**	**9.0**	**10.4**
$k = 2$	5.6	5.7	**6.6**	**5.9**	**5.8**	**7.6**
$k = 5$	5.0	5.1	**6.8**	5.1	**5.9**	**9.2**
$k = 10$	5.3	5.7	**6.6**	5.5	**7.0**	**11.2**
$k = 20$	5.1	5.3	**6.5**	5.3	**7.5**	**12.7**
$k = 50$	5.2	5.1	5.5	5.6	**8.1**	**16.9**
$k = 100$	4.9	5.1	4.8	5.6	**9.1**	**20.5**

Bold: above the 99% Monte Carlo band around 5% (4.21–5.79 for 5,000 searches); italic: below it.

Table 24: Experiment F (Null Zoo v2b): size and raw / size-adjusted power at a Sharpe ratio of 2 of every variant (percent).

	Size						Raw / size-adjusted power		
Family	RC	SPA_c-U	SPA_c (>)	SPA_c	SPA_c-BT	SPA_c-NB	RC	SPA_c-BT	SPA_c-NB
IID normal	5.3	5.5	**7.5**	**7.5**	5.5	*4.0*	53.7 / 52.4	53.7 / 51.8	46.9 / 50.3
Student t_4	4.6	4.8	**7.5**	**7.5**	5.3	*4.1*	51.4 / 52.0	53.9 / 52.9	47.4 / 50.2
Skew −1.32	5.2	5.5	**12.7**	**12.7**	5.5	*4.1*	55.8 / 55.0	48.1 / 46.0	41.8 / 44.1
Skew +1.32	5.1	5.1	*4.1*	*4.1*	5.5	*3.8*	51.4 / 51.0	60.9 / 59.5	54.4 / 58.7
GARCH(1,1)	4.8	5.0	**6.2**	**6.2**	4.8	*3.3*	50.6 / 51.1	53.8 / 54.1	48.6 / 54.2
AR(1), $\phi = 0.2$	**6.6**	**6.8**	**9.7**	**9.7**	**6.9**	5.5	39.7 / 34.3	39.9 / 33.8	33.5 / 31.6
Two vol. regimes	4.8	4.9	**6.0**	**6.0**	5.4	*3.4*	49.7 / 50.6	54.6 / 53.5	49.8 / 54.5
Correlated, $\rho = 0.5$	**5.7**	**5.7**	**7.0**	**7.0**	**5.7**	*4.3*	57.7 / 55.6	57.3 / 55.0	51.3 / 53.5
Clusters, $\rho = 0.8$	**5.6**	**5.7**	**7.1**	**7.1**	**5.6**	*4.4*	60.4 / 58.5	60.1 / 57.6	53.5 / 55.7

Bold: above the 99% Monte Carlo band around 5% (4.44–5.56 for 10,000 searches); italic: below it.

Table 25: Every window of the empirical illustration (p-values; Null Zoo v2b, part G).

	Mean block length 8					22		
Window	RC	SPA_c-U	SPA_c	SPA_c-BT	SPA_c-NB	SPA_c-U	SPA_c	LET
Mar 1927–Nov 1928[16]	0.980	0.979	1.000	1.000	1.000	0.991	1.000	1.000
Nov 1928–Jul 1930	0.212	0.212	0.221	0.175	0.180	0.182	0.198	0.934
Jul 1930–Apr 1932	**0.038**	**0.038**	**0.020**	**0.030**	**0.027**	**0.015**	**0.008**	0.148
Apr 1932–Dec 1933	0.755	0.755	0.754	0.765	0.762	0.761	0.754	1.000
Dec 1933–Aug 1935	0.622	0.622	0.611	0.621	0.620	0.638	0.620	1.000
Aug 1935–Apr 1937	0.734	0.725	0.647	0.746	0.714	0.759	0.591	1.000
May 1937–Jan 1939	0.289	0.289	0.270	0.287	0.287	0.271	0.275	0.970
Jan 1939–Sep 1940	0.267	0.267	0.245	0.258	0.254	0.278	0.262	0.956
Sep 1940–May 1942	0.113	0.113	0.119	0.099	0.119	0.066	0.067	0.555
May 1942–Jan 1944	0.828	0.828	0.830	0.830	0.831	0.848	0.850	1.000
Jan 1944–Oct 1945[14]	0.960	0.950	1.000	1.000	1.000	0.948	1.000	1.000
Oct 1945–Jul 1947	0.213	0.213	0.193	0.184	0.179	0.182	0.156	0.895
Jul 1947–Apr 1949	0.416	0.416	0.425	0.434	0.437	0.379	0.393	0.993
Apr 1949–Feb 1951	0.875	0.875	1.000	1.000	1.000	0.932	1.000	1.000
Feb 1951–Dec 1952	0.338	0.327	0.535	0.546	0.605	0.231	0.412	0.986
Dec 1952–Dec 1954	0.411	0.411	0.463	0.482	0.479	0.406	0.497	0.996
Dec 1954–Dec 1956	0.928	0.928	1.000	1.000	1.000	0.975	1.000	1.000
Dec 1956–Dec 1958	0.239	0.239	0.240	0.232	0.228	0.227	0.224	0.912
Dec 1958–Dec 1960	0.084	0.084	0.115	0.097	0.107	**0.032**	0.056	0.572
Dec 1960–Dec 1962	0.072	0.072	**0.047**	**0.050**	**0.024**	0.101	0.084	0.486
Dec 1962–Dec 1964	0.934	0.934	1.000	1.000	1.000	0.970	1.000	1.000
Dec 1964–Dec 1966	0.099	0.099	0.090	0.089	0.067	0.087	0.070	0.422
Dec 1966–Jan 1969	0.085	0.085	0.166	0.169	0.269	0.069	0.112	0.584
Jan 1969–Jan 1971	**0.019**	**0.019**	**0.010**	**0.007**	**0.003**	**0.022**	**0.016**	**0.008**
Jan 1971–Jan 1973	0.063	0.063	**0.049**	0.115	0.094	**0.031**	**0.032**	0.190
Jan 1973–Jan 1975	**0.010**	**0.010**	**0.000**	**0.004**	**0.003**	**0.015**	**0.001**	**0.022**
Jan 1975–Jan 1977	0.626	0.555	0.675	0.714	0.723	0.530	0.645	0.998
Jan 1977–Jan 1979	0.078	0.078	0.074	0.068	0.055	**0.042**	**0.048**	0.248
Jan 1979–Jan 1981	0.405	0.405	0.539	0.526	0.571	0.383	0.528	0.995
Jan 1981–Jan 1983	0.159	0.159	0.125	0.145	0.134	0.116	0.081	0.540
Jan 1983–Jan 1985	0.287	0.287	0.295	0.338	0.330	0.232	0.269	0.927
Jan 1985–Jan 1987	0.856	0.791	0.817	0.817	0.815	0.831	0.840	1.000
Jan 1987–Dec 1988	0.324	0.324	0.334	0.307	0.327	0.298	0.317	0.994
Jan 1989–Dec 1990	0.347	0.347	0.339	0.337	0.348	0.334	0.337	0.977
Dec 1990–Dec 1992	0.919	0.871	1.000	1.000	1.000	0.876	1.000	1.000
Dec 1992–Dec 1994	0.754	0.754	0.750	0.751	0.753	0.769	0.764	1.000
Dec 1994–Dec 1996	0.952	0.952	1.000	1.000	1.000	0.978	1.000	1.000
Dec 1996–Dec 1998	0.740	0.740	0.741	0.753	0.749	0.791	0.796	1.000
Dec 1998–Dec 2000	0.897	0.897	1.000	1.000	1.000	0.912	1.000	1.000
Dec 2000–Dec 2002	0.225	0.225	0.184	0.202	0.196	0.208	0.172	0.902
Dec 2002–Dec 2004	0.894	0.894	1.000	1.000	1.000	0.909	1.000	1.000
Dec 2004–Dec 2006	0.939	0.928	1.000	1.000	1.000	0.955	1.000	1.000
Dec 2006–Dec 2008	0.236	0.236	0.231	0.247	0.239	0.218	0.240	0.989
Dec 2008–Dec 2010	0.932	0.932	1.000	1.000	1.000	0.964	1.000	1.000
Dec 2010–Dec 2012	0.894	0.894	1.000	1.000	1.000	0.917	1.000	1.000
Dec 2012–Dec 2014	0.963	0.895	1.000	1.000	1.000	0.846	1.000	1.000
Dec 2014–Dec 2016	0.924	0.918	1.000	1.000	1.000	0.972	1.000	1.000
Dec 2016–Jan 2019	0.610	0.610	0.603	0.630	0.643	0.607	0.596	1.000
Jan 2019–Dec 2020	0.598	0.598	0.600	0.615	0.615	0.596	0.594	1.000
Jan 2021–Jan 2023	0.599	0.599	0.616	0.612	0.626	0.594	0.607	1.000
Jan 2023–Jan 2025	0.973	0.970	1.000	1.000	1.000	0.981	1.000	1.000

As in Table 11.

Table 26: The nine predictions registered before the v2 run (analysis/v2/prereg-v2.json, commit `7658b3b46ddb`) and whether each held, as scored by analysis/v2/evaluate_v2.py.

ID	Prediction	Outcome
P1	Replication: in every one of the nine families, the null size of spa_c and of rc in experiment A differs from v1b's spa_consistent and reality_check by at most three combined Monte Carlo standard errors.	held
P2	Variance noise: spa_c_sd has size at most 0.06 in iid_normal, student_t4, skew_positive, garch, regimes, correlated_trials and block_cluster (experiment A).	**did not hold**
P3	Skew: in skew_negative, spa_c_sd has size above 0.08 and spa_c_boot_t has size at most 0.065 (experiment A).	held
P4	Bootstrap-t: spa_c_boot_t has size at most 0.065 in all eight families other than ar1 (experiment A).	held
P5	arch 8.0.0: spa_c_unstud has size at most 0.065 in all eight families other than ar1 (experiment A), and in experiment D arch's p-values are identical under studentize=True and studentize=False on every search.	held
P6	Default block: in iid_normal, spa_c_b22 has size at least one percentage point above spa_c at block 8 (experiment A).	held
P7	Sample size and strategies: in iid_normal, the size of spa_c at n = 2520 is at least one percentage point below its size at n = 252, and at k = 100 at least one percentage point above its size at k = 5 (experiment B).	held
P8	Studentizing pays off: in heterogeneous_volatility, the size-adjusted power of spa_c_boot_t exceeds that of rc by at least 0.05; in poor_alternatives, the raw power of spa_c_boot_t exceeds that of rc by at least 0.05 (experiment C).	held
P9	Recentring: in poor_alternatives under the null, rc has size below 0.03 and spa_c_boot_t has size at most 0.065 (experiment C).	**did not hold**

Code names: `spa_c` = SPA_c; `spa_c_sd` = SPA_c-SD; `spa_c_boot_t` = SPA_c-BT; `rc` = RC; `spa_c_unstud` = SPA_c-U; `spa_c_b22` = SPA_c (22); `spa_u` = SPA_u.

Table 27: The seven predictions registered before the v2b run (analysis/v2b/prereg-v2b.json, commit `ea2d963e3825`) and whether each held, as scored by analysis/v2b/evaluate_v2b.py.

ID	Prediction	Outcome
P10	Replication on a new grid: in experiment E at k = 20 and block 8, spa_c_ge has size at least 0.065.	held
P11	The bootstrap's own variance: in experiment E at k = 1, rc's size at block 22 exceeds its size at block 1 by at least one percentage point.	held
P12	Many strategies dampen it: in experiment E at block 8, rc's size at k = 1 exceeds its size at k = 100 by at least half a percentage point.	**did not hold**
P13	Block re-studentization: in experiment F, spa_c_nb_ge has size at most 0.065 in all eight families other than ar1.	held
P14	Under autocorrelation: in experiment F's ar1 family, spa_c_nb_ge's size is at least half a percentage point below spa_c_boot_t_ge's.	held
P15	Ranking by mean helps when the skilled strategy is the most volatile: in experiment C2 with the skill on strategy 19, rc's raw power is at least spa_c_boot_t_ge's.	held
P16	Counting ties: in experiment C2's poor alternatives, spa_c_boot_t_ge has size at most 0.065 and raw power at least 0.20 above rc's.	held

Code names: `rc` = RC; `spa_c_ge` = SPA_c; `spa_c_strict` = SPA_c (>); `spa_c_boot_t_ge` = SPA_c-BT; `spa_c_nb_ge` = SPA_c-NB.

Table 28: The 15 hypotheses registered before the confirmation run v2c (analysis/confirm/prereg-confirm.json, commit `c0996a70714f`) and their outcomes, as scored by analysis/confirm/evaluate_confirm.py. Each holds only if every one of its checks does.

ID	Hypothesis	Checks	Outcome
H1	arch882 at block 8: size ≤ 0.065 in each of the nine families.	9/9	held
H2	arch882 at its default block 22: size ≤ 0.070 in each of the nine families.	9/9	held
H3	On the same searches at block 22, the fixed-variance spa_c's size exceeds arch882's by at least 0.020 in each of the nine families.	9/9	held
H4	The author's corrected consistent SPA test: size ≤ 0.065 in each of the six families without cross-sectional or serial correlation (iid_normal, student_t4, skew_negative, skew_positive, garch, regimes).	6/6	held
H5	On the same searches, the author's corrected consistent SPA test and the Reality Check differ in size by at most 0.010 in each of the nine families.	9/9	held
H6	Negative skew: the v1 implementation's consistent SPA test and the fixed-variance spa_c at block 8 each have size ≥ 0.10, while the corrected implementation and arch882 at block 8 each have size ≤ 0.065.	4/4	held
H7	The corrected StepM (any strategy declared superior at 5%) and the corrected consistent SPA test differ in size by at most 0.005 in each of the twelve joint-test designs (nine families, three autocorrelation levels).	12/12	held
H8	LET and the one-sided haircut, which ignore autocorrelation, have size above the upper edge of the 99% band around 0.05 (0.0556 for 10,000 searches) at each of phi = 0.02, 0.05, 0.1.	6/6	held
H9	Proposition 2: for LET and the one-sided haircut at each of phi = 0.02, 0.05, 0.1, the simulated size is within three Monte Carlo standard errors (computed at the closed-form size) of closed-form.json's value.	6/6	held
H10	LET-Lo and the one-sided haircut-Lo have size within the 99% band around 0.05 at each of phi = 0.02, 0.05, 0.1.	5/6	**did not hold**
H11	At each of phi = 0.02, 0.05, 0.1: arch882 at block 8, the Reality Check at block 8, the corrected SPA test and the corrected StepM each have size $\leq$ 0.065, and arch882 at block 22 has size ≤ 0.070.	15/15	held
H12	At each of phi = 0.02, 0.05, 0.1: the fixed-variance spa_c at block 22 has size ≥ 0.09 and the v1 implementation's consistent SPA test (block 8) has size ≥ 0.065.	6/6	held
H13	At phi = 0.05 and 0.1, each of the four block-8 tests of H11 (arch882, Reality Check, corrected SPA, corrected StepM) has size at least 0.010 below the one-sided haircut's simulated size at the same phi.	8/8	held
H14	Equicorrelated extension of Proposition 2: for LET and the one-sided haircut in each of the six (rho, phi) cells, the simulated size is within three Monte Carlo standard errors (computed at the closed-form size) of closed-form.json's equicorrelated value.	12/12	held
H15	At rho = 0.6, for LET and the one-sided haircut at each phi = 0, 0.05, 0.1, the simulated size is more than three Monte Carlo standard errors (computed at the simulated size) below Proposition 2's independent-strategy value: the independence formula overstates the size of correlated searches.	6/6	held

arch882: arch at the merge commit of pull request 882; spa_c: the fixed-variance SPA test; corrected: the author's implementation after canlicapital PR 562; v1: before it. Every check, with its value and bar, is in analysis/confirm/evaluation-confirm.json.